



Electromagnetic Compatibility (EMC) and Immunity (EMI) of Power Electronic Circuits



Electromagnetic Compatibility (EMC)

Electromagnetic compatibility (EMC) describes the ability of a device, system, or piece of equipment to work properly in its electromagnetic environment without causing problems for other devices nearby. A system is considered electromagnetically compatible if it satisfies three basic conditions: it does not interfere with other systems, it is not negatively affected by interference coming from other systems, and it does not interfere with its own operation.

To make sure these conditions are met, governments and international organizations establish EMC standards. These standards define how much electromagnetic emission is acceptable and how resistant a device must be to external interference. The specific standards that apply depend on where the product is manufactured or used.

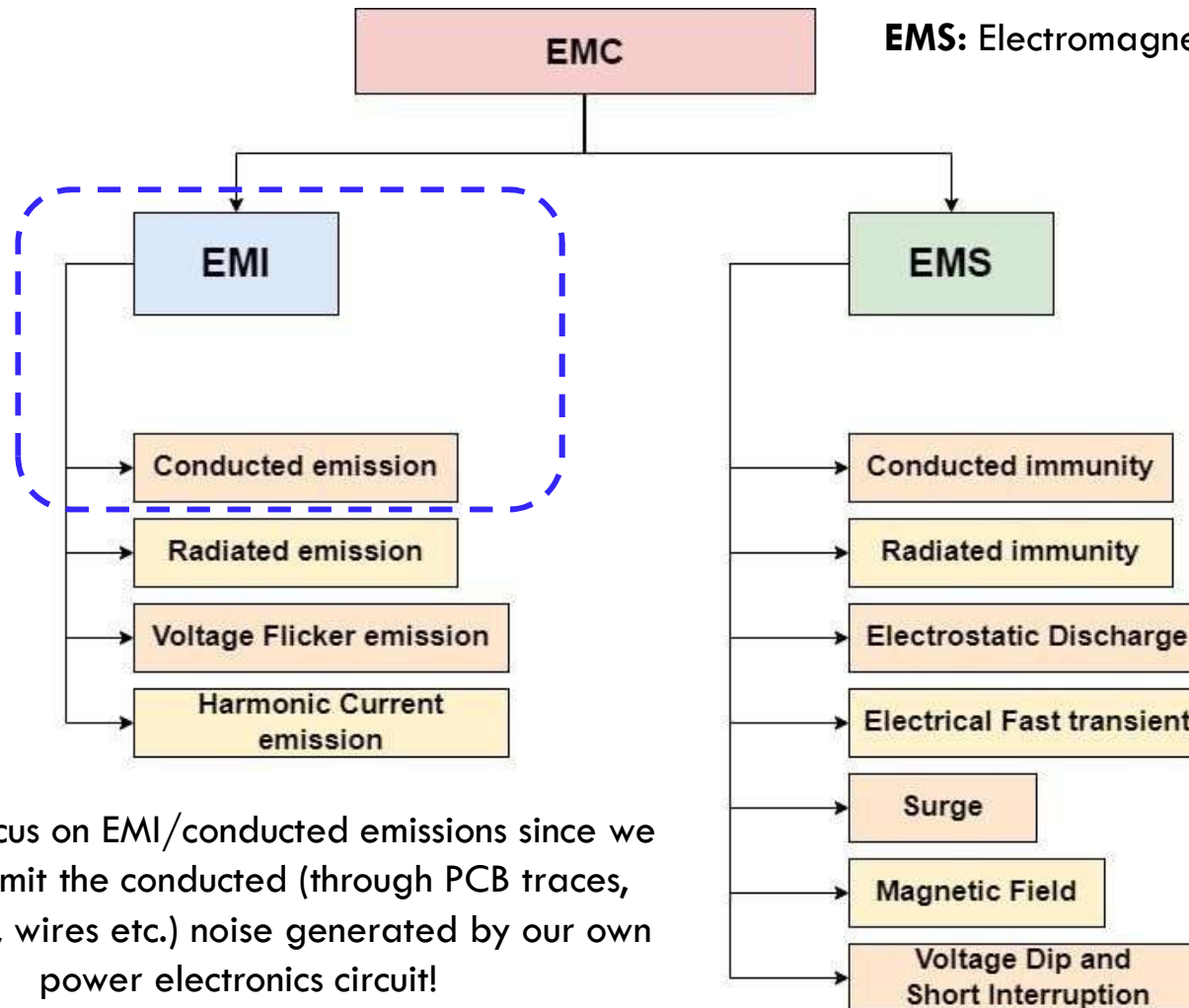
At the international level, EMC standards are developed and published by the International Electrotechnical Commission. The main EMC-related standardization groups under the IEC are IEC TC 77 and CISPR. These standards define measurement methods, test procedures, and allowable emission limits to ensure electromagnetic compatibility.



Electromagnetic Compatibility (EMC)

EMI: Electromagnetic Interference

EMS: Electromagnetic Susceptibility/Immunity



We will focus on EMI/conducted emissions since we want to limit the conducted (through PCB traces, connectors, wires etc.) noise generated by our own power electronics circuit!

Electromagnetic Interference (EMI)

Electromagnetic emissions can be categorized into two main types: **conducted emissions** and **radiated emissions**. Conducted emissions enter an electronic system through power supply lines and cables. In contrast, radiated emissions arise from electromagnetic waves generated by power and signal lines, switching components, and electrostatic discharge events. These emissions propagate through free space from electronic devices and PCB traces, potentially causing interference in nearby systems.

Typical examples include mobile phones or laptops disturbing aircraft electronic equipment.

Conducted interference can be reduced by placing line filters close to power inputs or connector interfaces. Another commonly used technique for suppressing conducted noise is the application of ferrite cores or ferrite rings. Ferrite cores attenuate high-frequency noise by dissipating high-frequency currents within a ferrite ceramic material, thereby acting as effective noise suppression elements.

Electromagnetic emissions can also originate from high-frequency signal traces. In addition, inadequate decoupling of power and ground planes can lead to unwanted emissions. Such conditions may give rise to unintended current components, including common-mode (CM) and differential-mode (DM) currents (will be shown later).

Electromagnetic Interference (EMI)

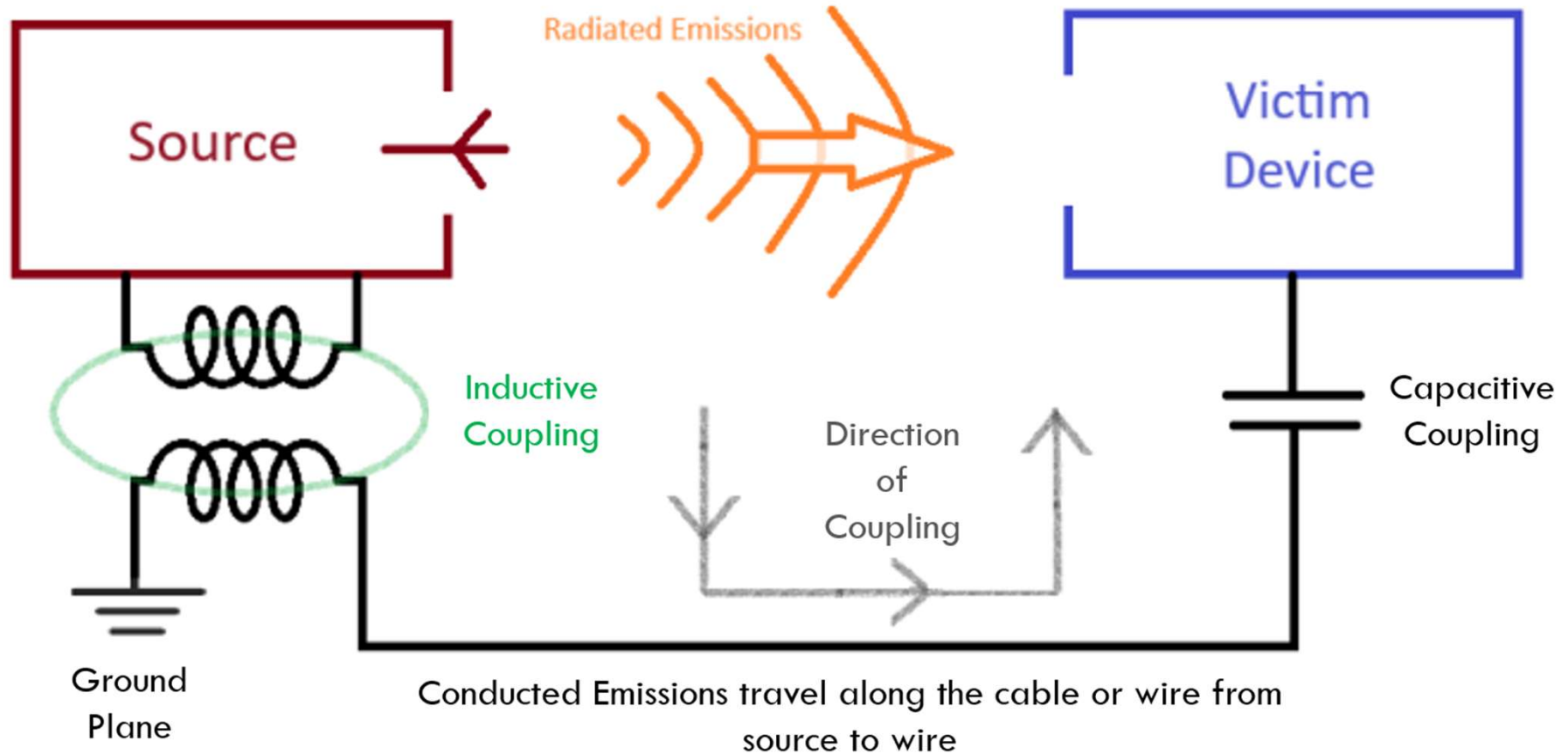
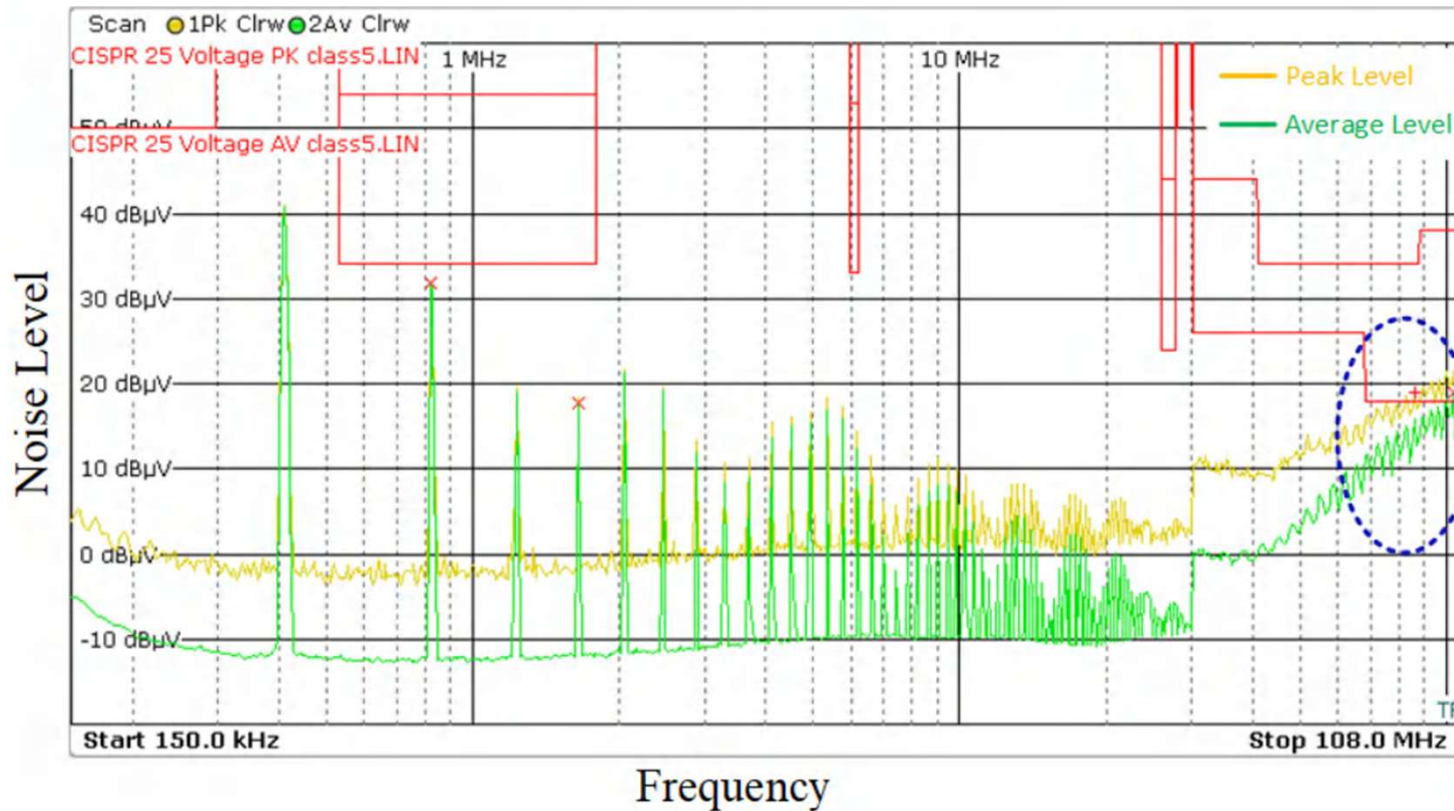


Diagram illustrating some of the possible coupling paths of Electromagnetic Interference (EMI)

Electromagnetic Interference (EMI)

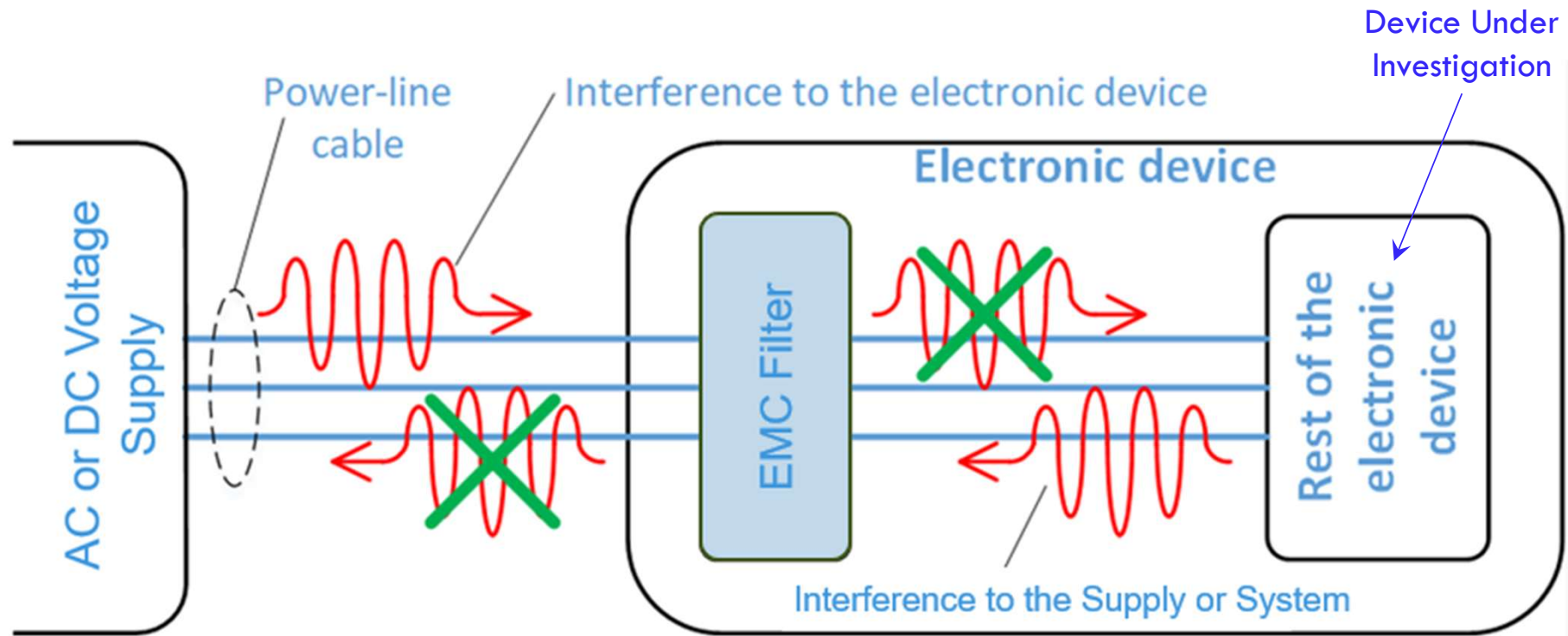
Conducted Emission Test of a Buck Converter according to CISPR 25 Standard



$$\text{dB}\mu\text{V} = 20 \log_{10} \left(\frac{V}{1 \mu\text{V}} \right)$$
$$\text{dB}\mu\text{V} = 20 \log_{10}(V) + 120$$

Violation of EMC Specification

Electromagnetic Interference (EMI)

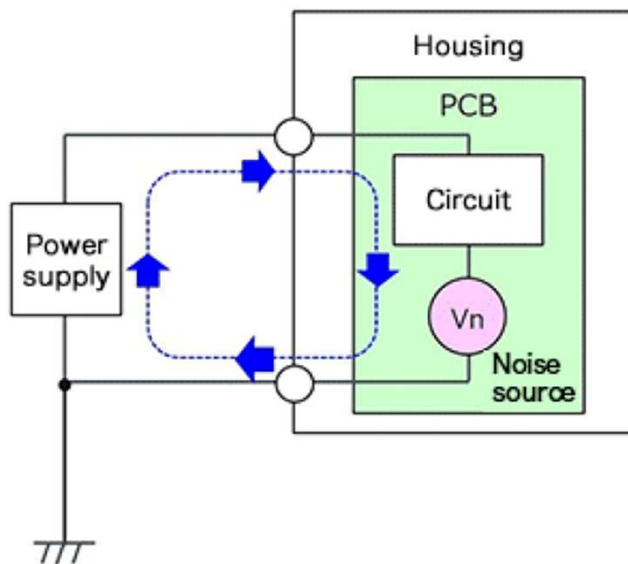


EMC Filter filters in both directions

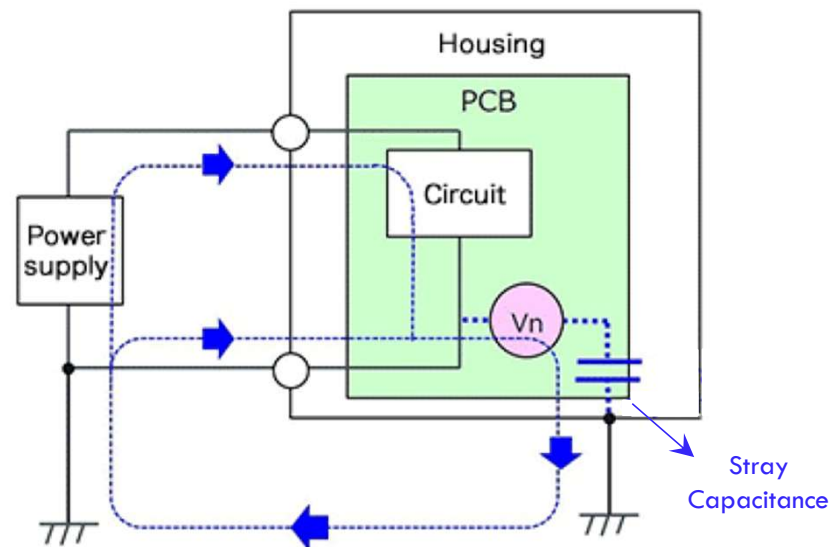
Electromagnetic Interference (EMI)

Conducted emissions can be divided into two main categories. The first type is differential-mode noise. The second type is common-mode noise. These two noise mechanisms are commonly explained using circuit diagrams. In this discussion, the focus is on power supply systems, where the diagrams represent practical examples of electronic circuits implemented on printed circuit boards (PCBs) housed within an enclosure, with electrical power supplied from an external source.

Differential-Mode Noise



Common-Mode Noise





Electromagnetic Interference (EMI)

In **differential-mode noise**, the noise source is present between the power supply lines and appears in series with the supply. The resulting noise current flows in the same direction as the normal power supply current. It is referred to as differential-mode noise because the forward and return currents flow in opposite directions.

Common-mode noise occurs when a noise current leaks through stray capacitances or similar paths, flows through ground, and then returns to the power supply. It is called common-mode noise because the noise currents on both the positive (+) and negative (−) power supply lines flow in the same direction. In this case, no noise voltage appears directly across the power supply lines.



As explained above, these noise types are classified as conducted emissions. However, because noise currents flow through the power supply lines, they can also give rise to radiated emissions.

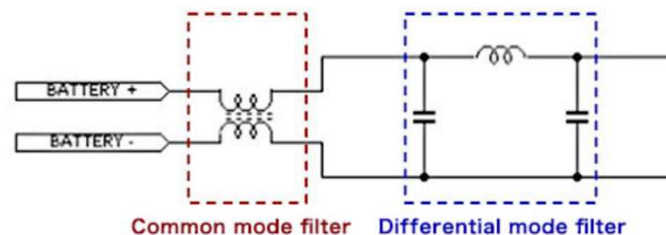
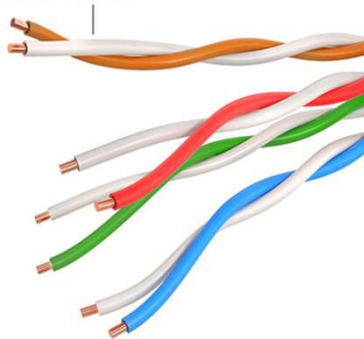


Electromagnetic Interference (EMI)

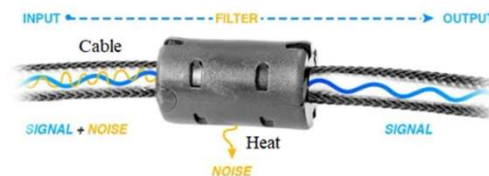
A key takeaway from these calculation results is that, for identical noise current levels, radiation caused by common-mode noise is significantly higher—about 100 times greater in this example. Whenever conducted or radiated emissions that contribute to EMI exceed the permissible limits, appropriate noise mitigation measures must be applied. In particular, when dealing with radiated emissions, countermeasures against common-mode noise are especially critical.

Noise countermeasures will be discussed in sequence, but the most fundamental approaches are as follows: for differential-mode noise, the loop area S should be reduced, for instance by using twisted-pair cables; for common-mode noise, the cable length should be minimized as much as possible. In practice, however, constraints related to component placement and component selection often arise. Therefore, additional techniques, such as the use of filters, must also be considered.

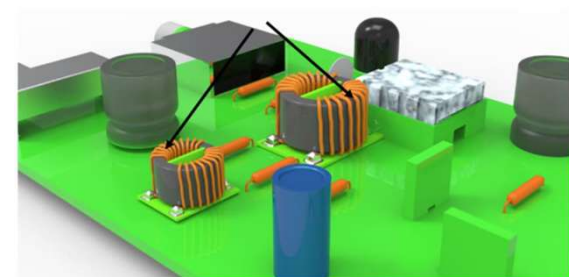
Twisted Pairs



EMC CM- and DM-Filters

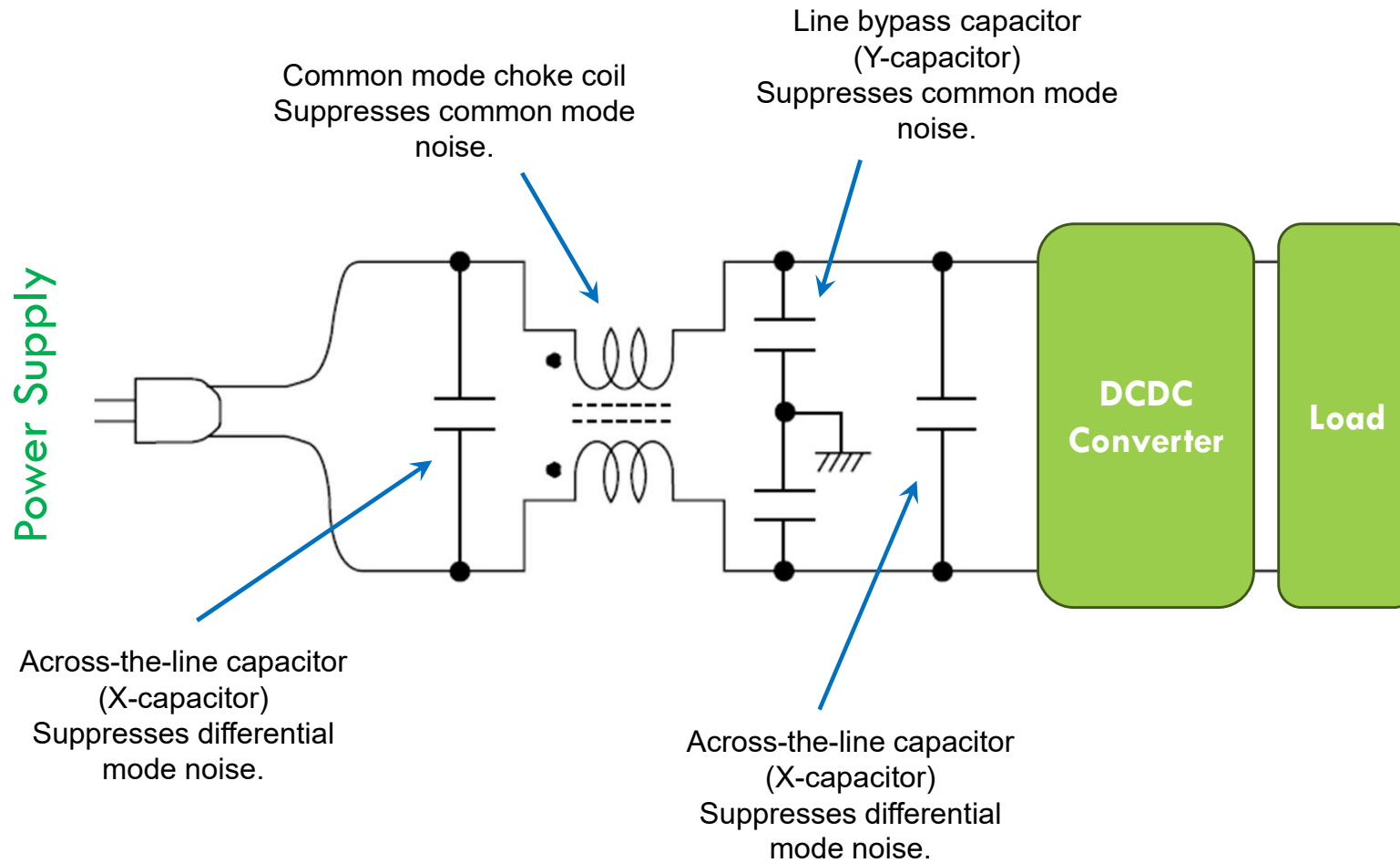


Ferrite Clamp for CM-Noise Suppression on a Cable



DM- and CM-Filter
Components mounted on a
PCB

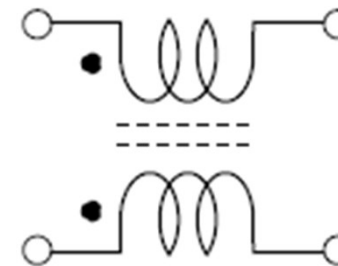
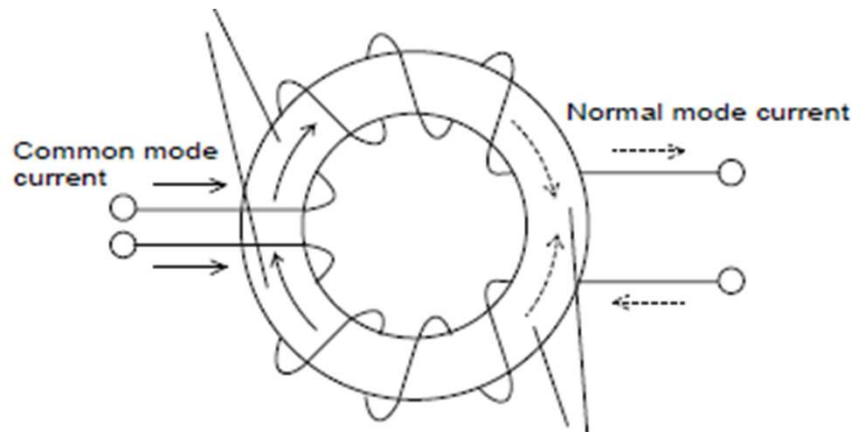
EMC Filter



EMC Filter

Common mode choke coils work as a simple wire against differential mode current (signal), while they work as an inductor against common mode current (noise).

Magnetic flux caused by common mode current is accumulated, producing impedance.



Equivalent Circuit

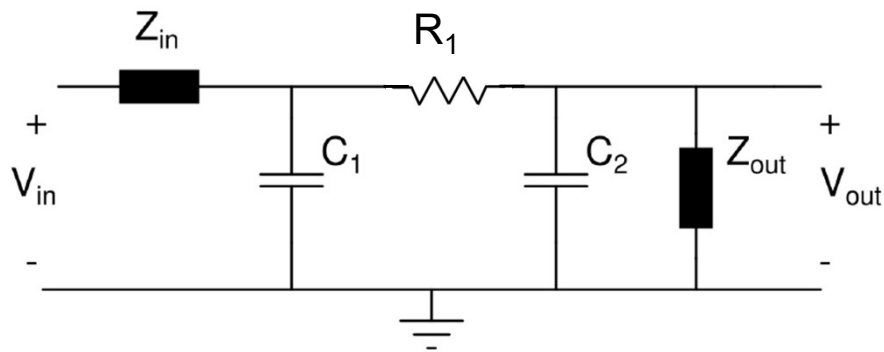
Magnetic flux caused by differential mode current cancels each other, and impedance is not produced.

Since magnetic flux caused by common mode current is accumulated, a high amount of impedance is produced.

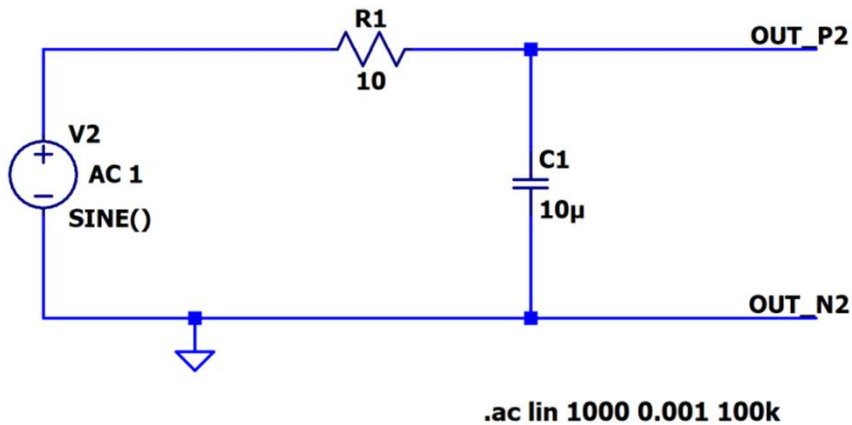
Common mode choke coils are suited for common mode noise suppression because a coil with large impedance is easily achieved.

RC Filter – Differential Filtering

Example: Differential-Mode Filter with PI-Topology



LTSpice Simulation:



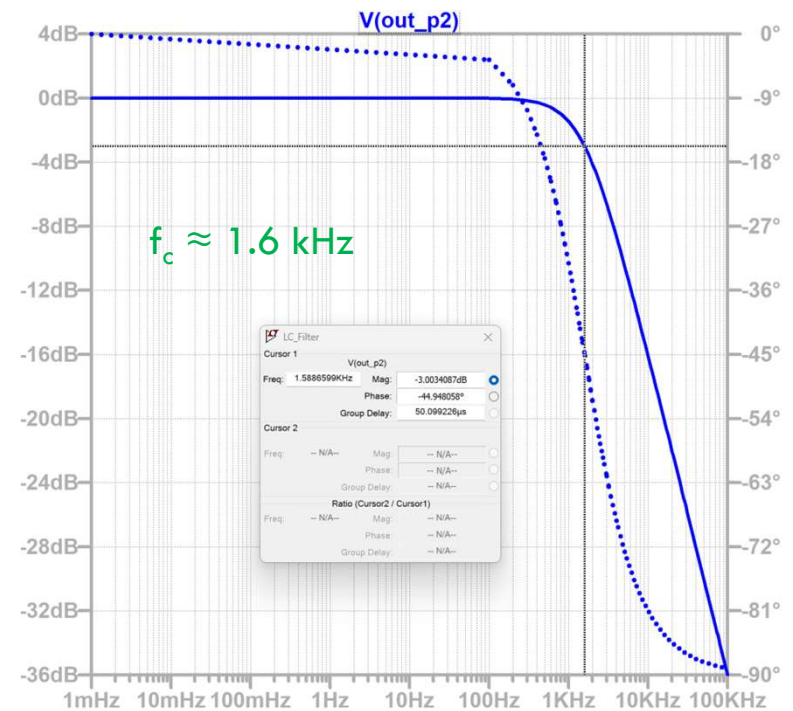
If we neglect Z_{in} and C_1 :

Transfer Function:

$$Z_C = \frac{1}{sC}$$

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{1 + sR_1C_2}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi R_1 C_2}$$

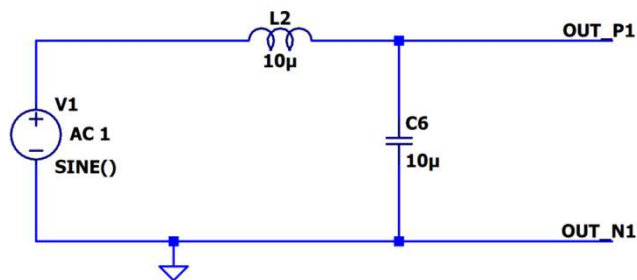
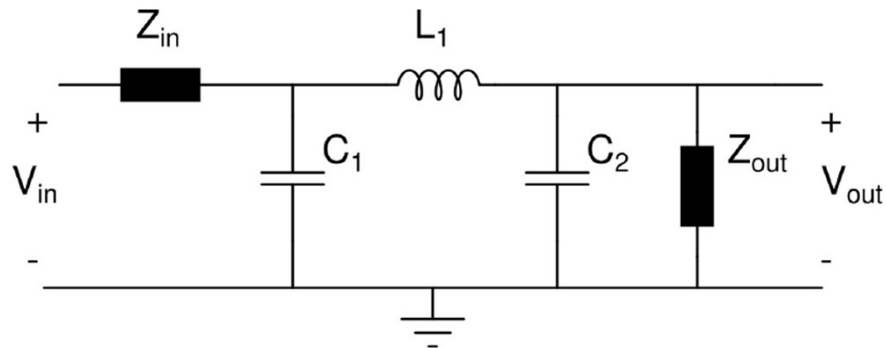


LC Filter – Differential Filtering

We cannot use an RC filter if we want to filter the supply voltage due to the voltage drop! RC filters are suitable for signal filtering, not for supply filtering!

Alternative: LC Filter!

Example: Differential-Mode Filter with PI-Topology



$$Z_C = \frac{1}{sC}$$

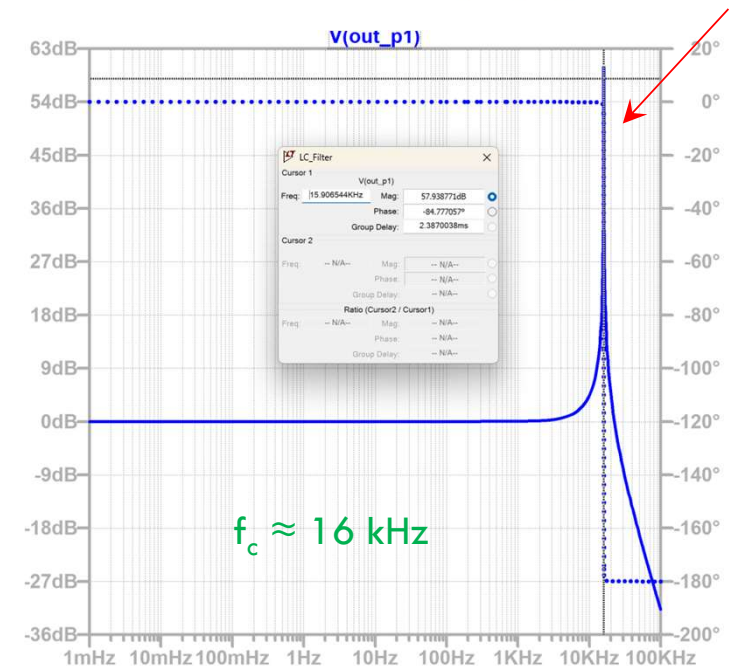
$$Z_L = sL$$

Transfer Function:

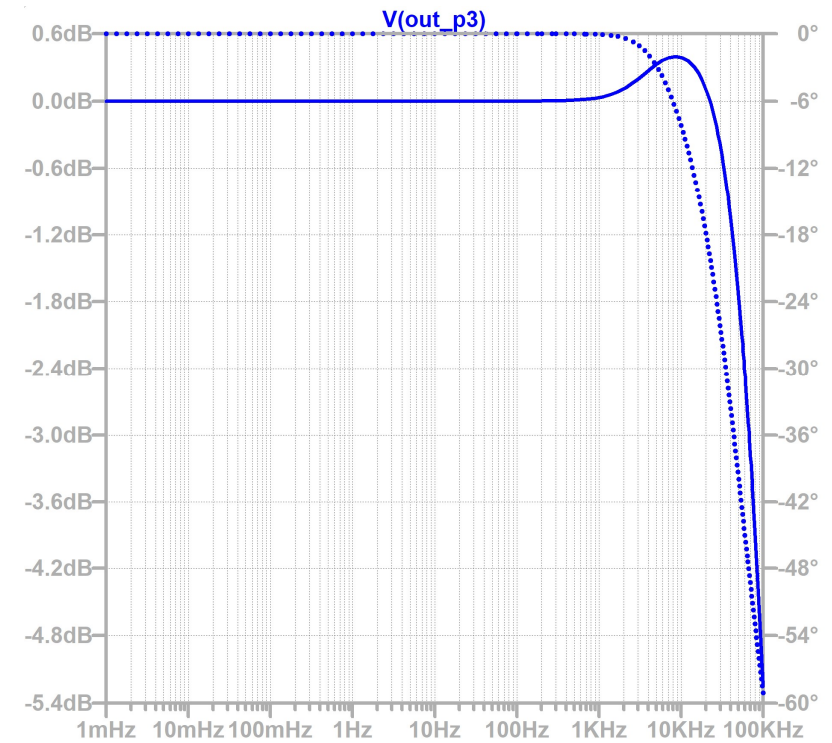
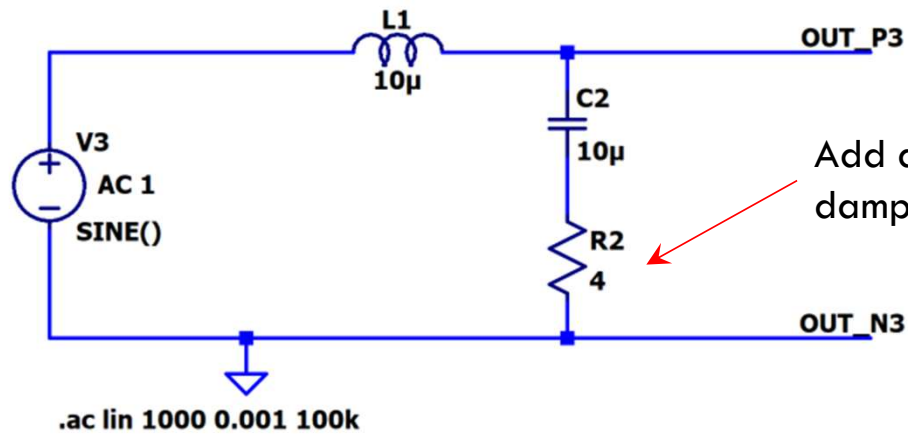
$$\frac{V_{out}}{V_{in}} = \frac{1}{1 + s^2 L_1 C_2}$$

$$f_c = \frac{1}{2\pi\sqrt{L_1 C_2}}$$

Resonance Frequency

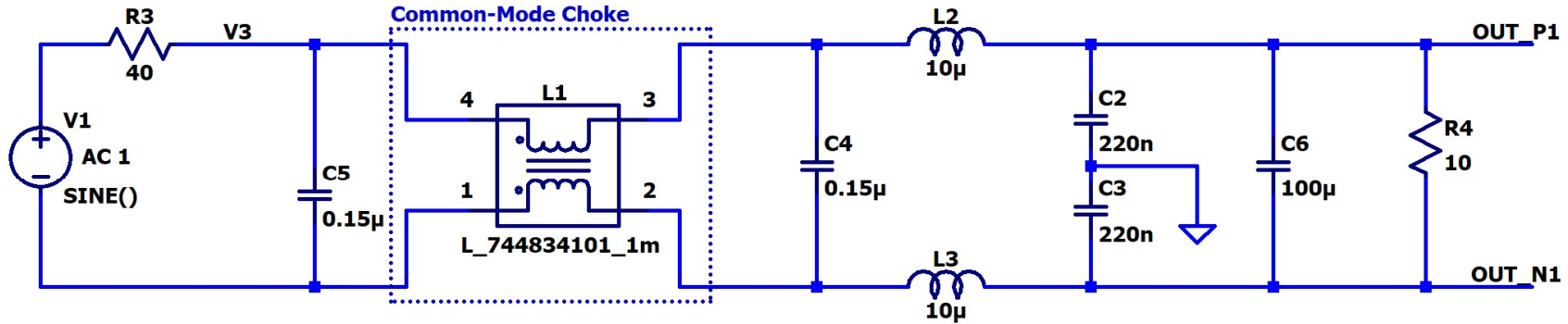


LC Filter – Differential Filtering

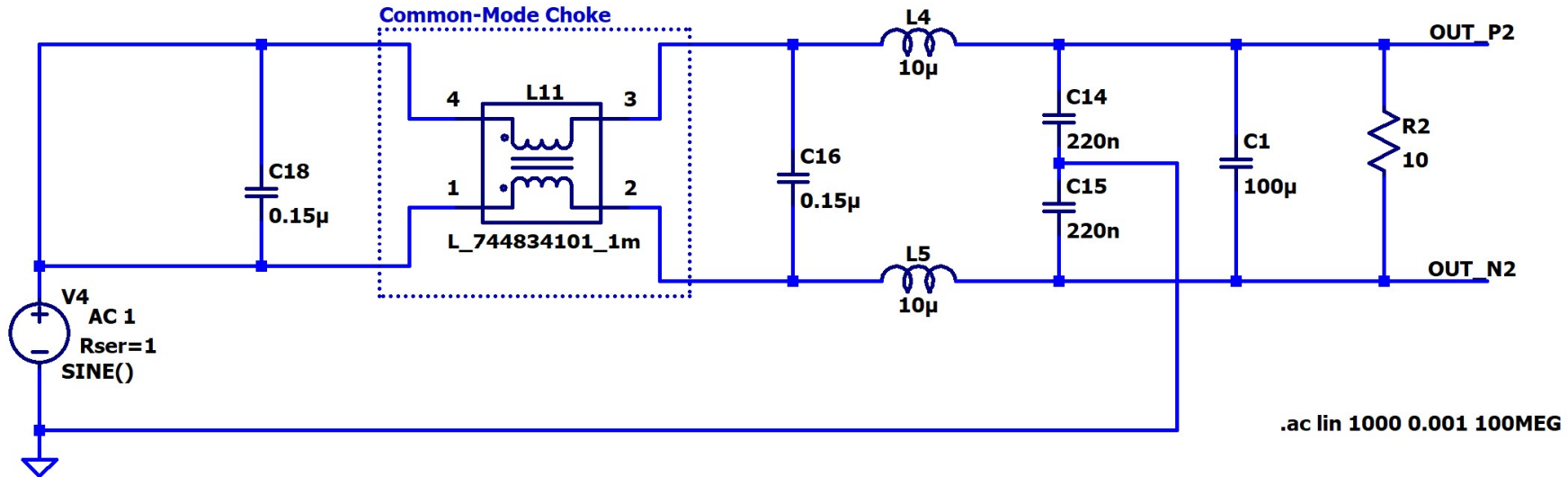


EMI Filter

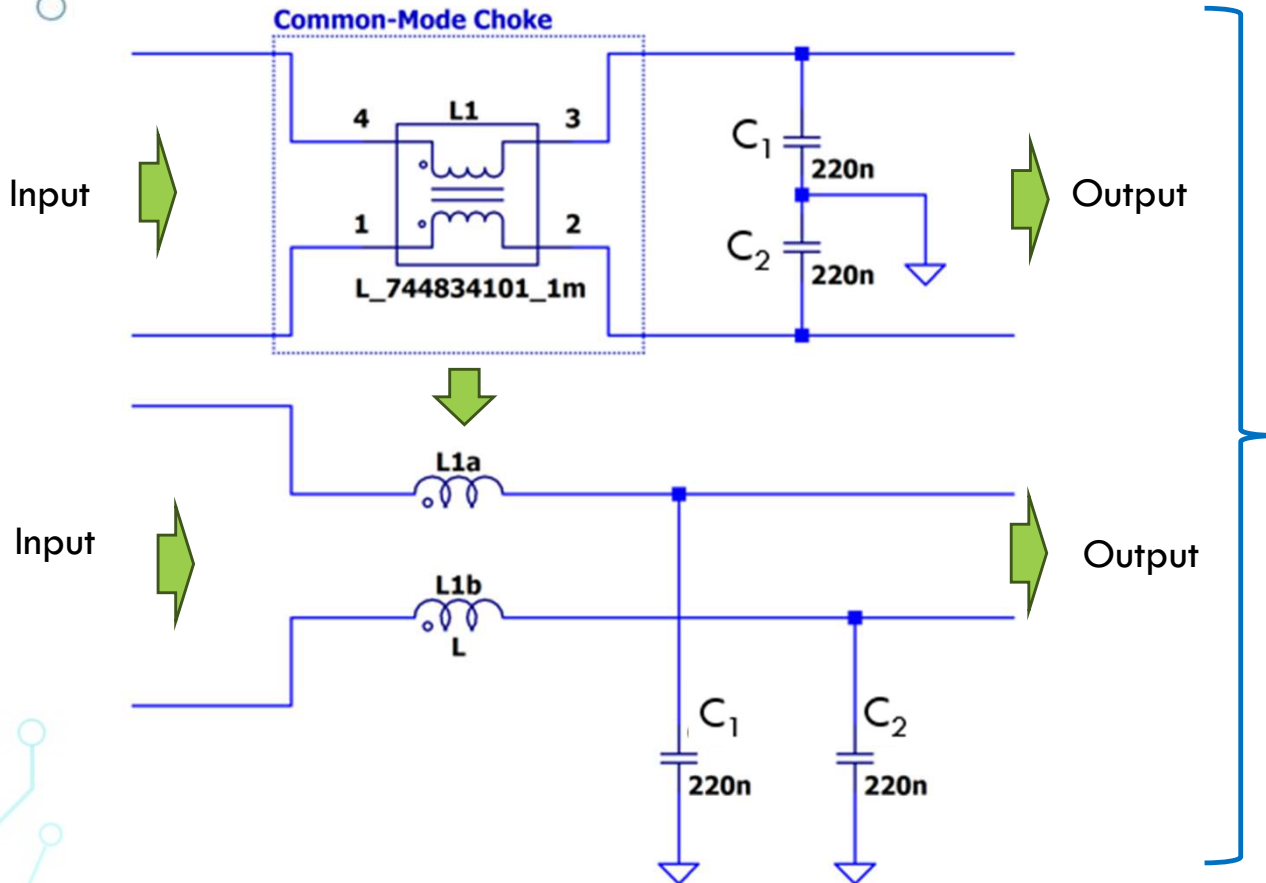
Differential-Mode Transfer Function



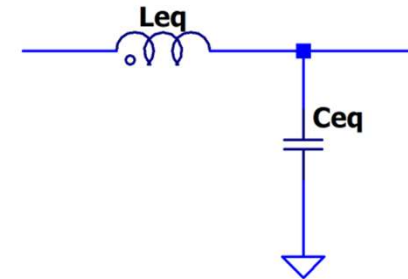
Common-Mode Transfer Function



EMI Filter



For common-mode signals:



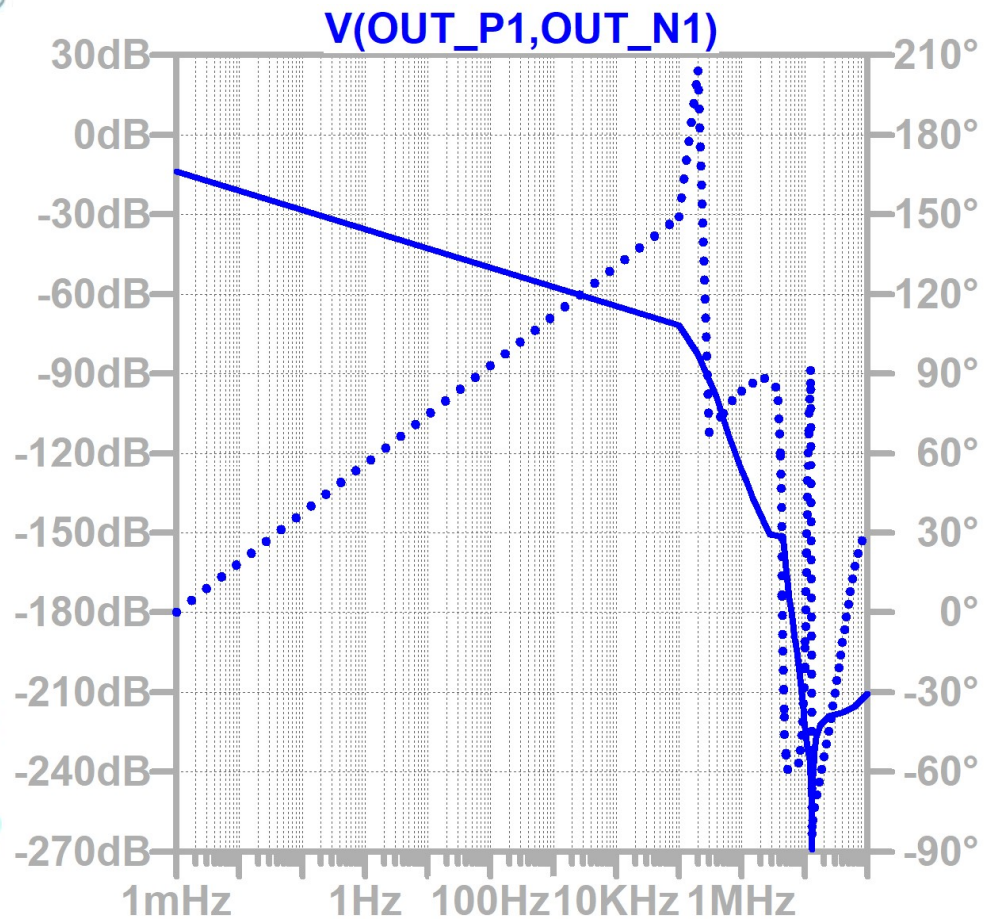
$$L_{eq} = L_{1a} + L_{1b} + k\sqrt{L_{1a} \cdot L_{1b}}$$

k : mutual coupling factor

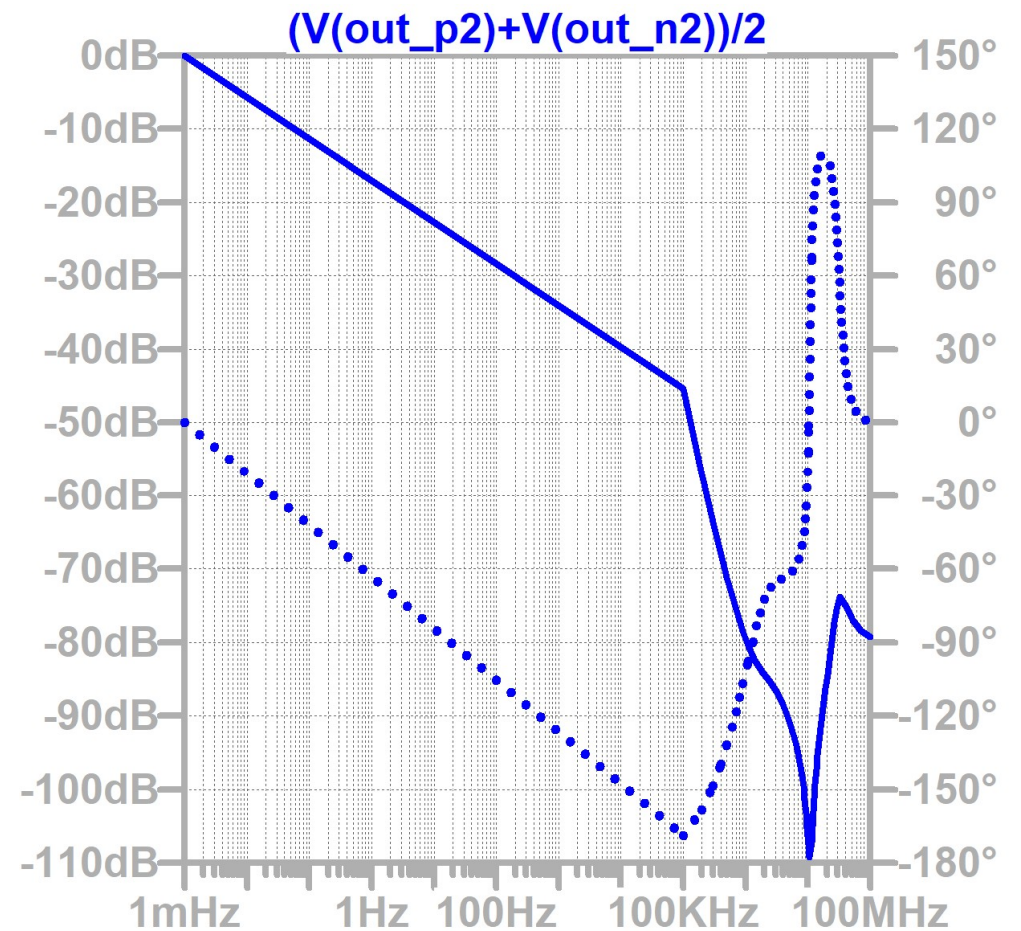
$$C_{eq} = C_1 + C_2$$

EMI Filter

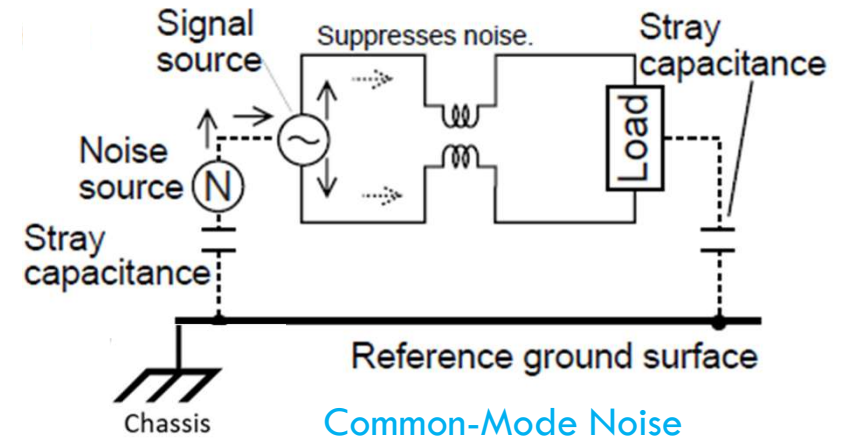
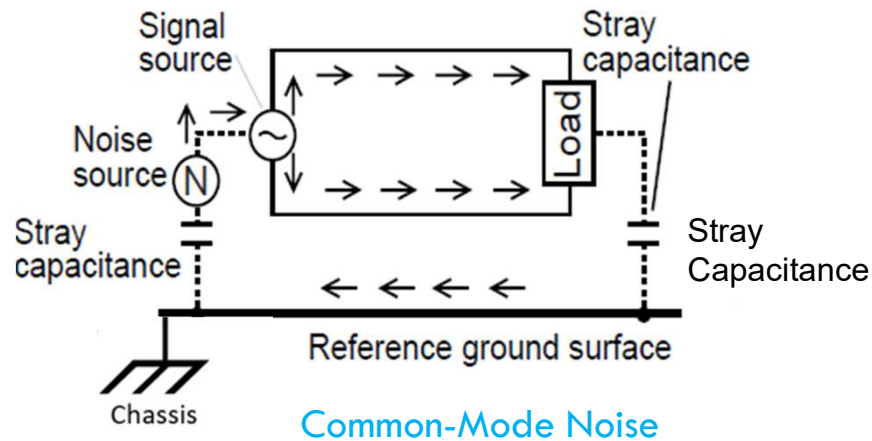
Differential Transfer Function



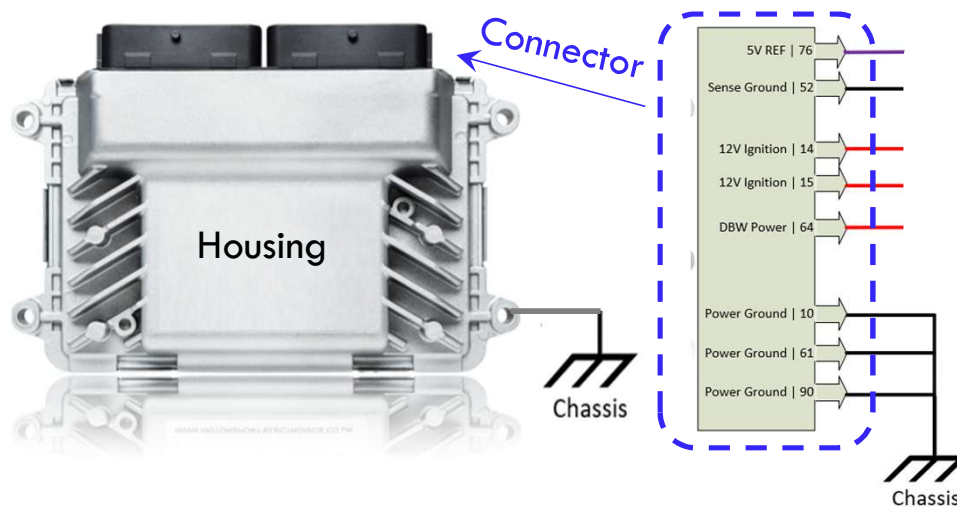
Common-Mode Transfer Function



Differential-Mode (DM) and Common-Mode (CM) Noise



Example: ECU in a Vehicle



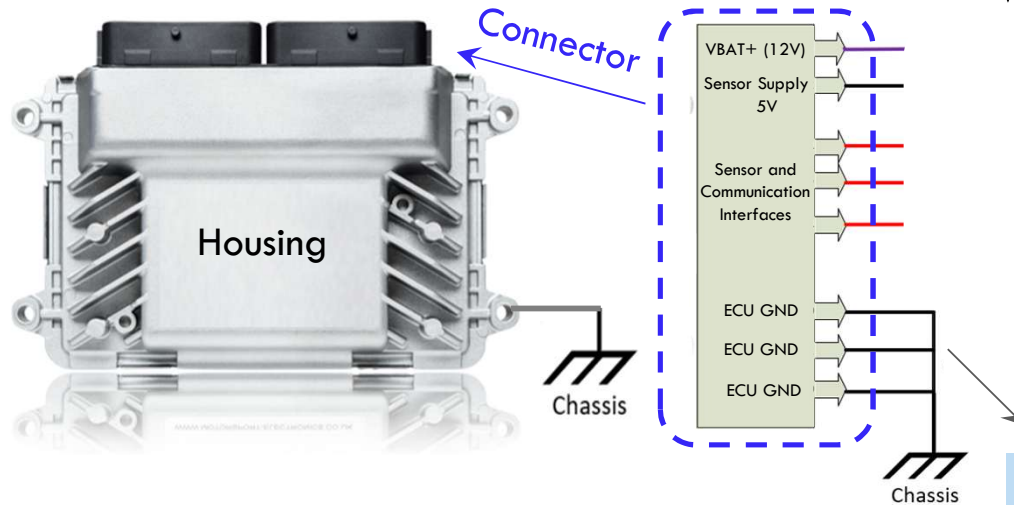
Vehicle Chassis



ECU ground connected to Vehicle Chassis (global ground) both through housing and connector)

Differential-Mode (DM) and Common-Mode (CM) Noise

Example: ECU in a Vehicle

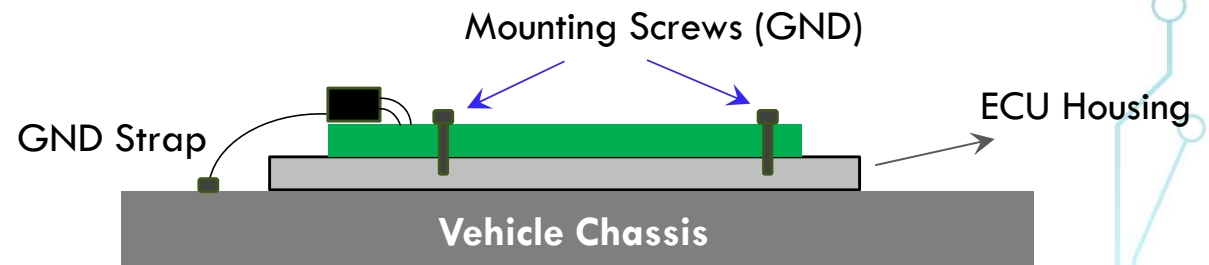
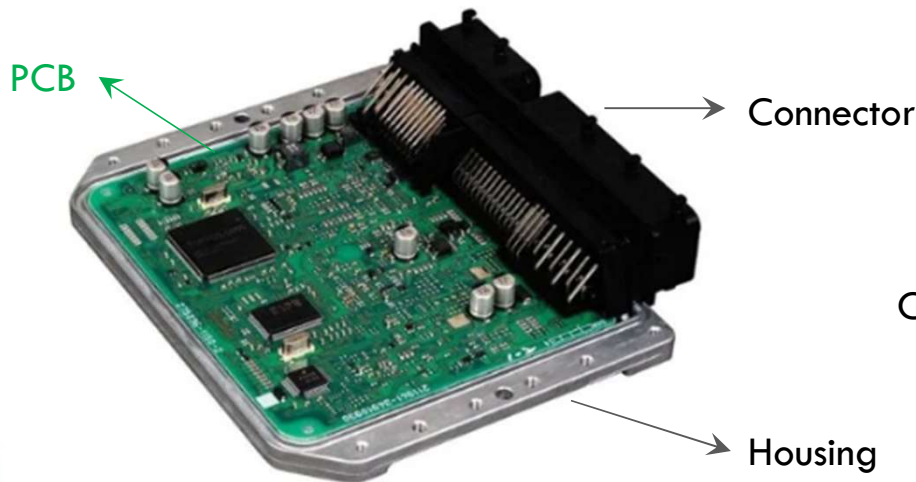


Vehicle Chassis

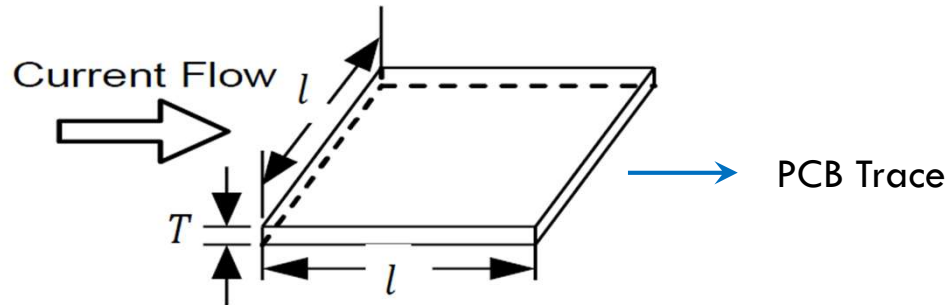


ECU ground connected to Vehicle Chassis (global ground) both through housing (mounting screws) and connector (cables)

Cable connects the connector ground to chassis



PCB Stray Resistances



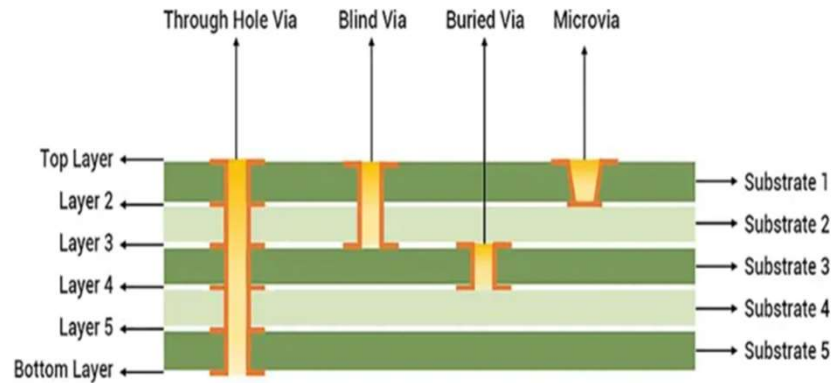
$$R = \frac{\rho \cdot l}{T \cdot l} = \frac{\rho}{T}$$

Material	mΩ-cm	mΩ-in
Copper	1.70	0.67
Gold	2.2	0.87
Lead	22.0	8.66
Silver	1.5	0.59
Silver (plated)	1.8	0.71
Tin -lead	15	5.91
Tin (plated)	11	4.33

Resistivity for Different Conductor Materials

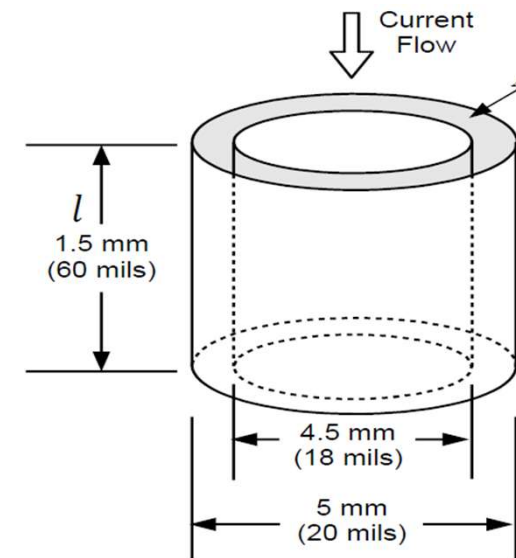
Copper Weight (Oz.)	Thickness (μm/mils)	mΩ per square (25°C)	mΩ per square (125°C)
1/2	17.5/0.7	1	1.4
1	35/1.4	0.5	0.7
2	70/2.8	0.25	0.35

PCB Stray Resistances



PCB Cross-Section with Vias

Vias have also stray resistances (and inductances)!

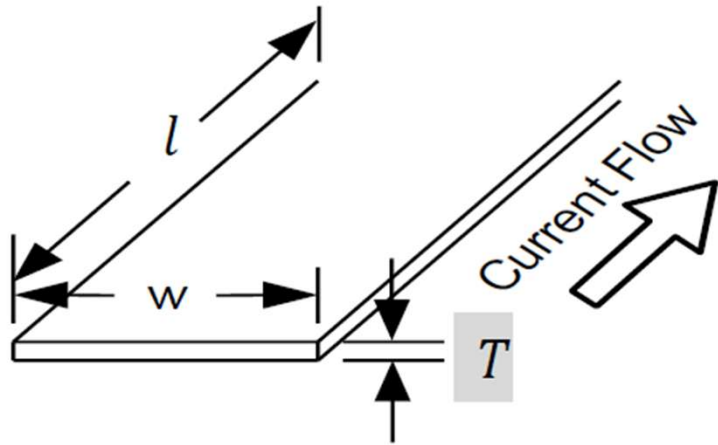


$$R = \frac{\rho \cdot l}{A}$$

$$R = \frac{\rho \cdot l}{\pi \cdot (r_o^2 - r_i^2)}$$

$$R = \frac{(0.67 \mu\Omega \cdot \text{in}) \cdot (0.06 \text{ in})}{\pi \cdot [(0.01 \text{ in})^2 - (0.009 \text{ in})^2]} = 0.67 \text{ m}\Omega$$

PCB Stray Inductances

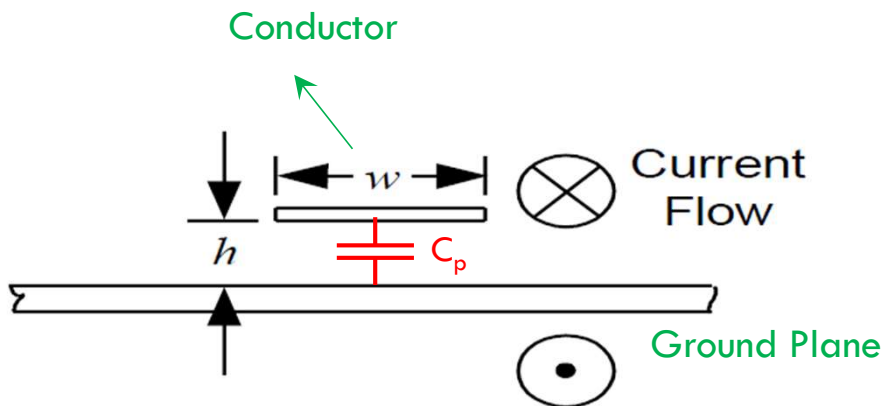


$$L = 2 \cdot l \cdot \left(\ln \left(\frac{l}{T + W} \right) + \frac{1}{2} \right) nH/cm$$

$$L = 5 \cdot l \cdot \left(\ln \left(\frac{l}{T + W} \right) + \frac{1}{2} \right) nH/in$$

W (mm/in)	T(mm/in)	Inductance (nH/cm or nH/in)
0.25/0.01	0.07/0.0028	10/24
2.5/0.1	0.07/0.0028	6/14
12.5/0.5	0.07/0.0028	2/6

PCB Traces over Ground Plane



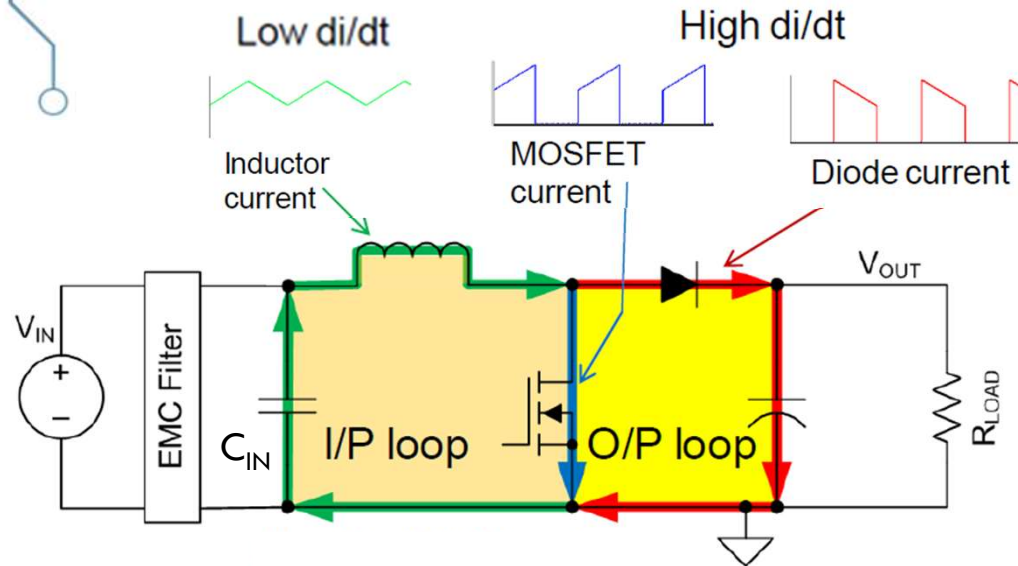
$$L = \frac{2 \cdot h \cdot l}{w} nH/cm$$

$$L = \frac{5 \cdot h \cdot l}{w} nH/in$$

Metric			English		
h (cm)	w (cm)	Inductance (nH/cm)	h (in)	w (in)	Inductance (nH/in)
0.25	2.5	0.2	0.01	0.1	0.5
1.5	2.5	1.2	0.06	0.1	3.0

Inductance of the PCB trace with a ground plane below is reduced to a parasitic capacitance (C_p) between the PCB trace and the ground plane!

Impact of Stray Inductances on Switching Converters



Input Loop:

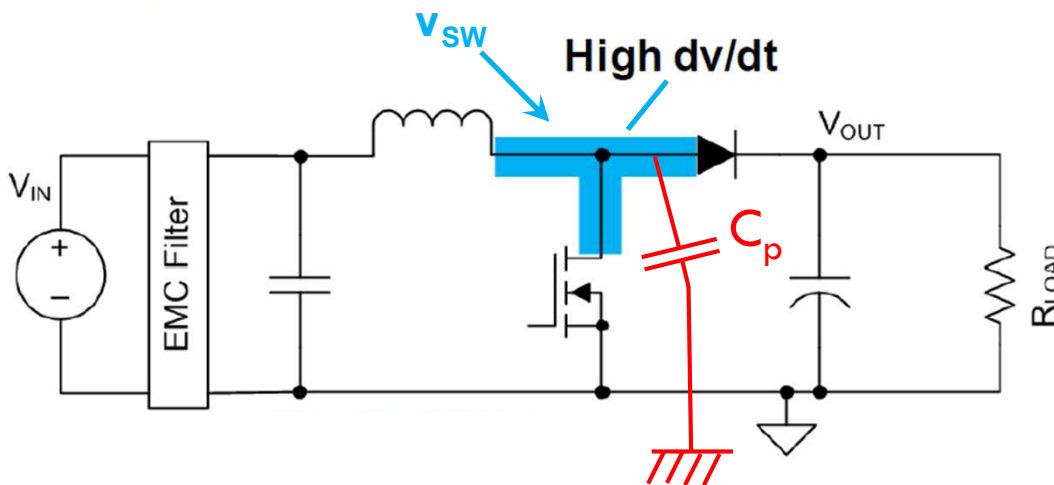
di/dt rates much lower

- Stray inductance is less critical
- C_{IN} provides local low impedance source

Output Loop:

Inductor current alternates between MOSFET and diode paths

- Pulsating currents
- Stray inductance will cause voltage spikes



High dv/dt at switched node (blue):

Switched between 0V and V_{OUT}

- Stray capacitance can cause:
 - EMI problems
 - Noise injected to internal circuits via heat sink

Minimize V_{SW} surface area!

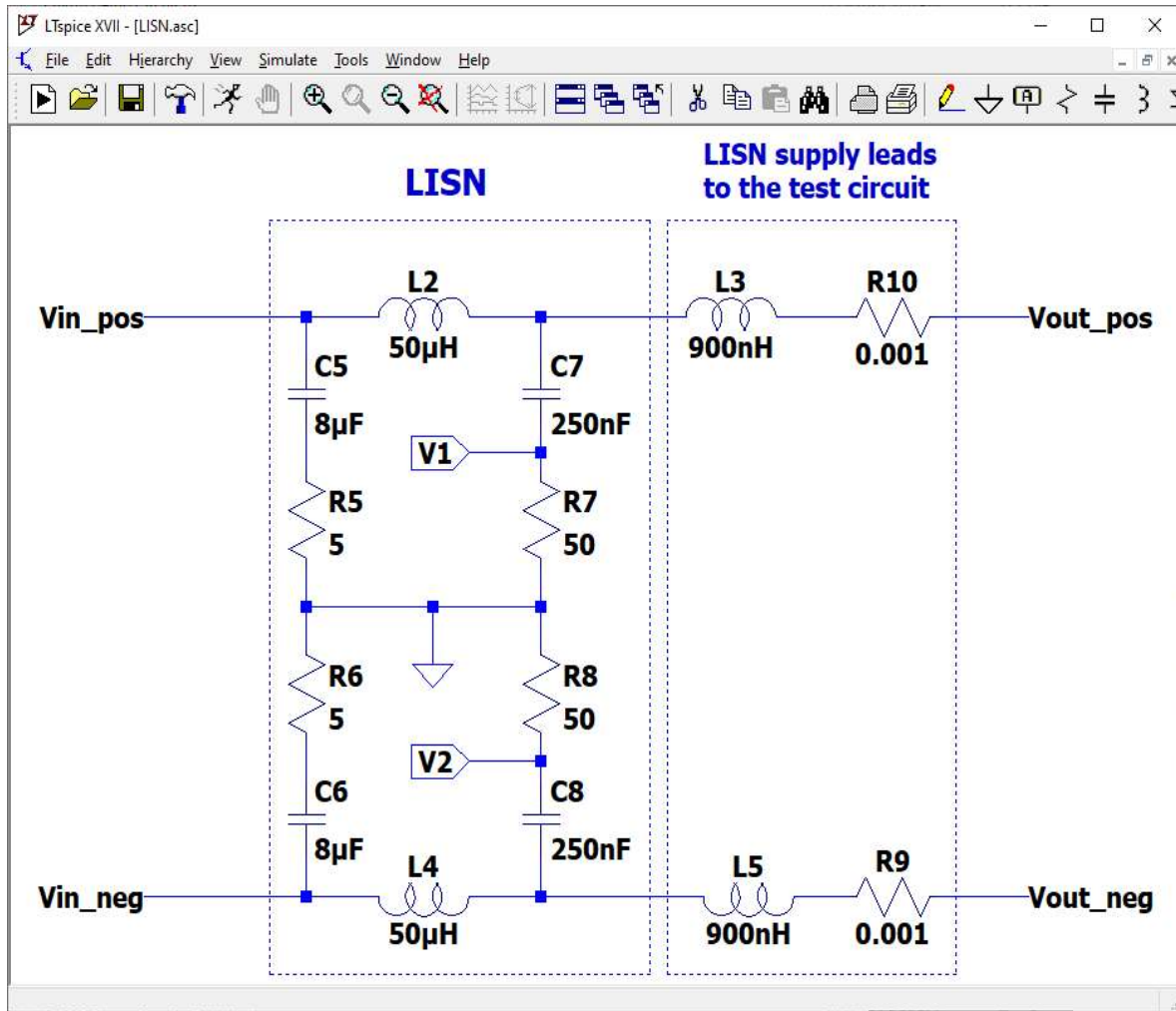


Example – Buck Converter

Buck converters play a key role in modern electronic systems, delivering efficient DC-DC power conversion in applications ranging from consumer electronics to industrial equipment. However, despite their high efficiency and widespread use, they introduce a major challenge: conducted emissions. The high-frequency switching inherent to their operation produces noise that can travel along power lines, causing electromagnetic interference (EMI). This interference may degrade the performance of the converter itself and negatively affect nearby electronic devices. Therefore, effective control and mitigation of these emissions are essential to ensure regulatory compliance and reliable system operation.



Example – Buck Converter

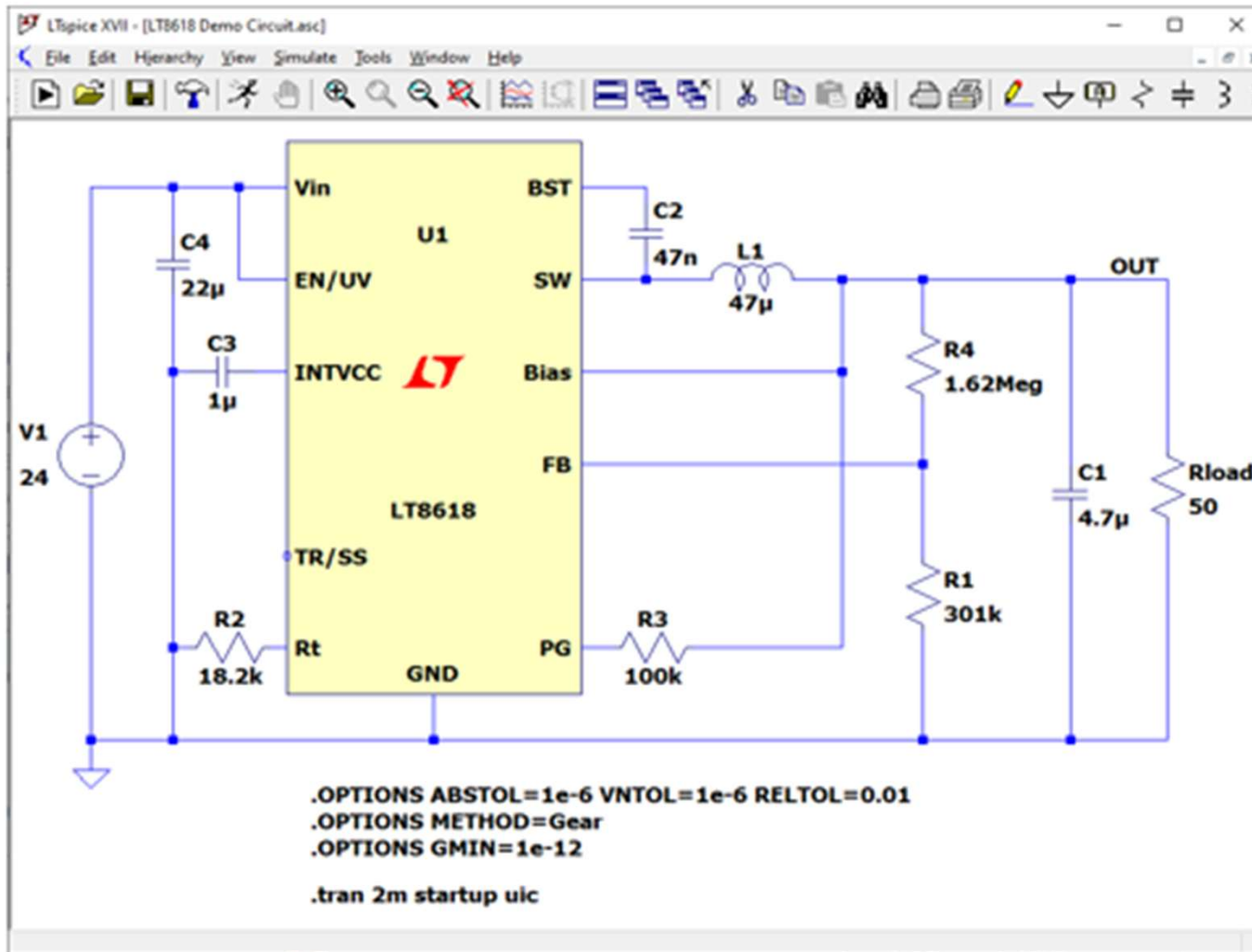


This circuit is the LTspice LISN circuit used in this article.

A Line Impedance Stabilization Network (LISN) is a test fixture used in EMC (Electromagnetic Compatibility) testing to provide a stable and repeatable impedance to the device under test (DUT) and to measure the conducted emissions on the power lines.

When simulating conducted emissions for a buck converter in LTspice, incorporating a LISN model can help predict the emissions and evaluate mitigation strategies

Example – Buck Converter



Circuit Description:

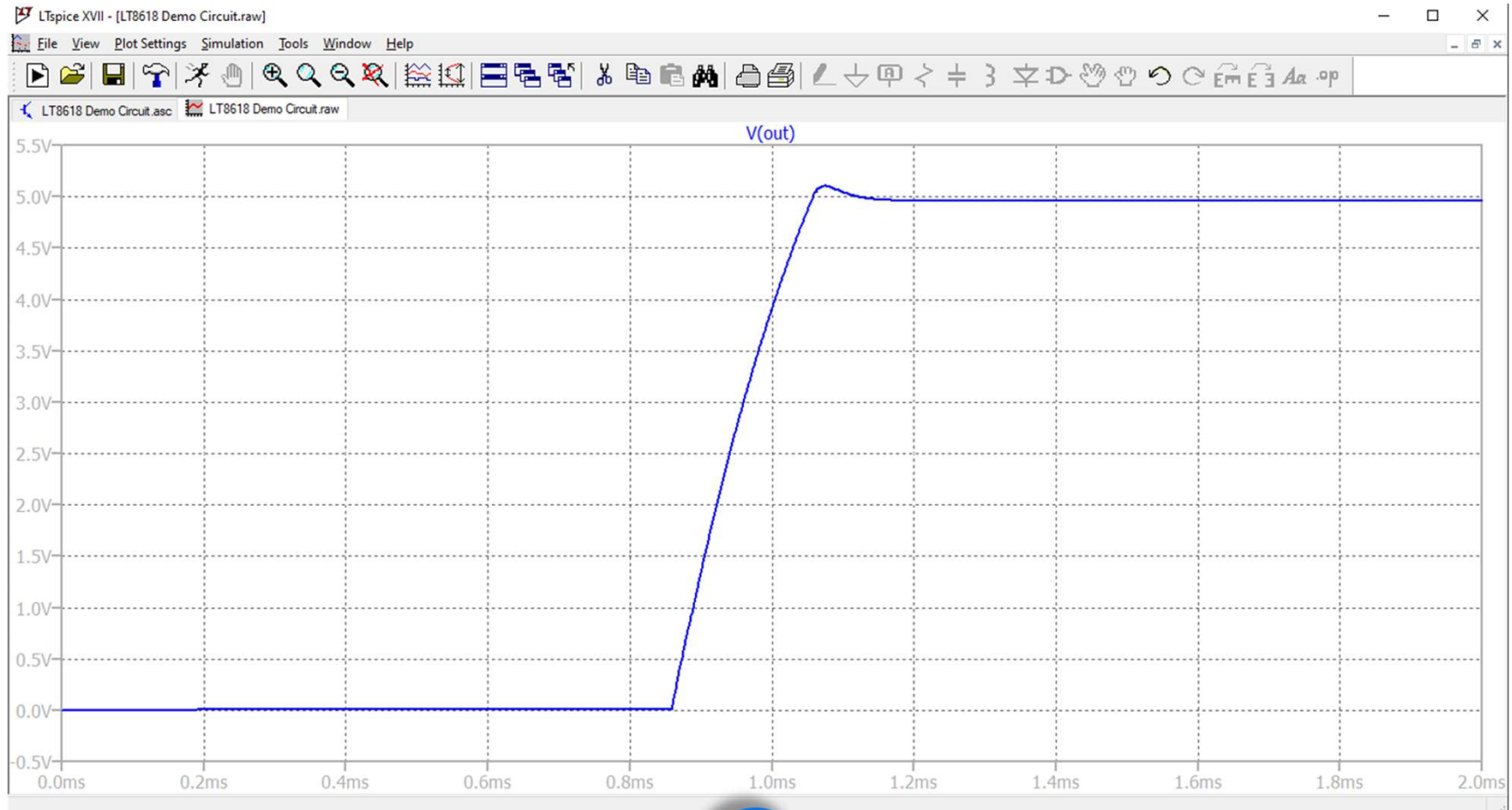
Buck (step-down) converter based on LT8618 chip.

Input: 24V

Output: 5V @ 100mA

F_{sw}: 2MHz

Example – Buck Converter





Example – Buck Converter

Performing a Fast Fourier Transform (FFT) in LTspice allows you to analyze the frequency content of a signal, which is particularly useful for examining harmonics, noise, and other spectral characteristics in your circuits. In our case the FFT will be used to analyze the Common Mode (CM) Noise.

Here's a step-by-step guide on how to perform an FFT in LTspice:

Step 1: Set Up Your Circuit

Ensure that the circuit is properly configured and that you have defined a transient analysis to capture the signal you want to analyze.

Step 2: Run the Transient Simulation

Run a transient analysis by including the `.TRAN` directive in your schematic. This analysis will generate the time-domain data needed for the FFT.

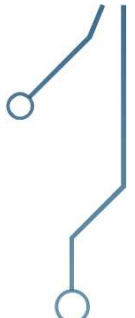
Step 3: View the Time-Domain Waveform

After running the simulation, view the time-domain waveform by clicking on the nodes of interest. In the circuits shown in the article, you would click on V1 and V2 to view the voltage across the capacitor.





Example – Buck Converter



Step 3: View the Time-Domain Waveform

After running the simulation, view the time-domain waveform by clicking on the nodes of interest. In the circuits shown in the article, you would click on V1 and V2 to view the voltage across the capacitor.

Step 4: Perform the FFT

Open the FFT Window: Right-click on the waveform viewer and select "View" -> "FFT" from the context menu.

Step 5. Select Waveforms to include in FFT

In the FFT window, select the signals to analyze. Ensure you have the correct waveforms selected (V(v1) and V(v2)).

Step 6: Select Visible Waveforms

In the windows select both signals V(v1) and V(v2)

Step 7: Create the Algebraic Expression for Common Mode (CM) Noise

Use the expression editor and create the algebraic expression: $(V(v1)+V(v2))*0.5*1000000$





Example – Buck Converter

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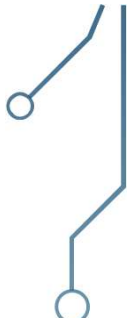
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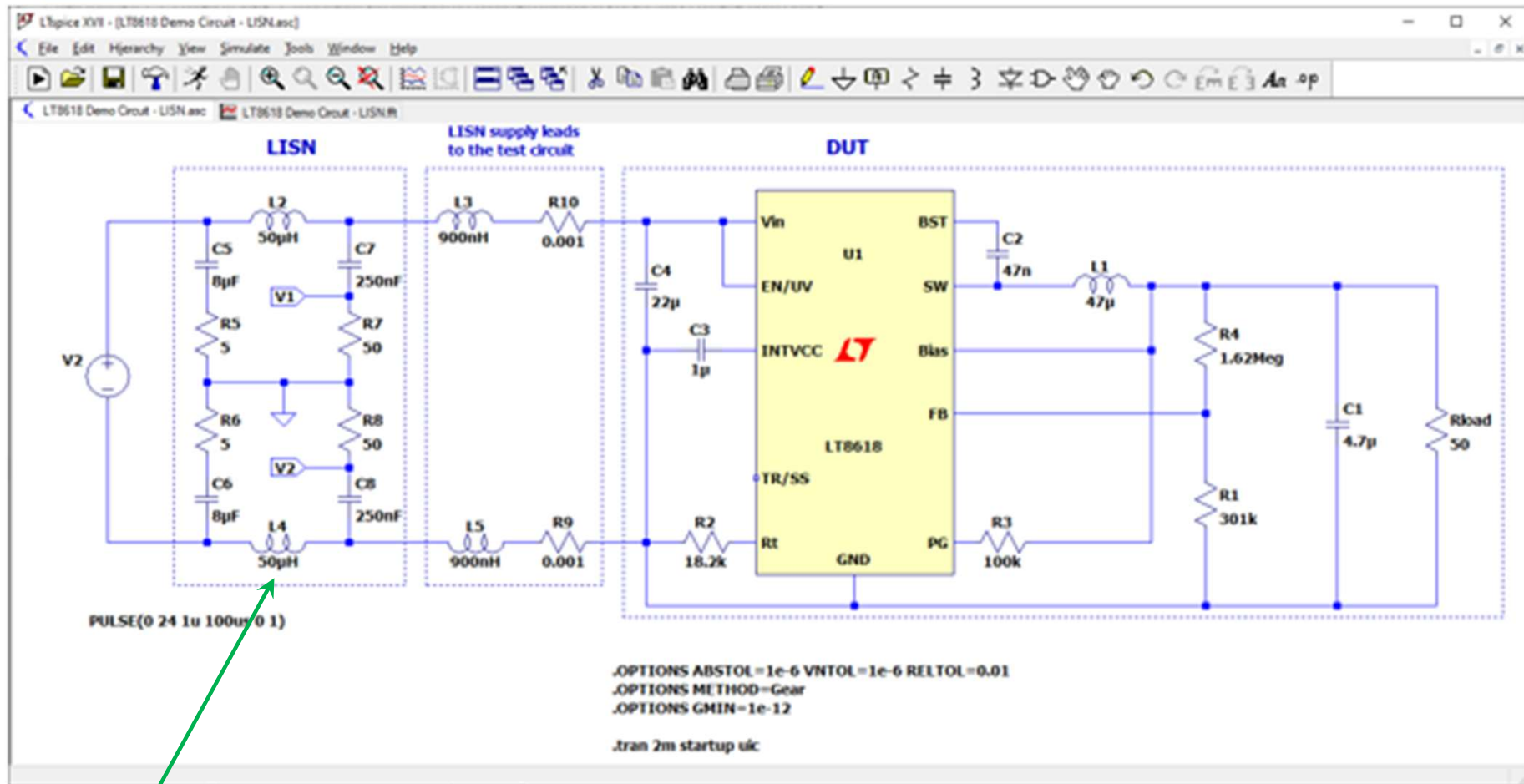
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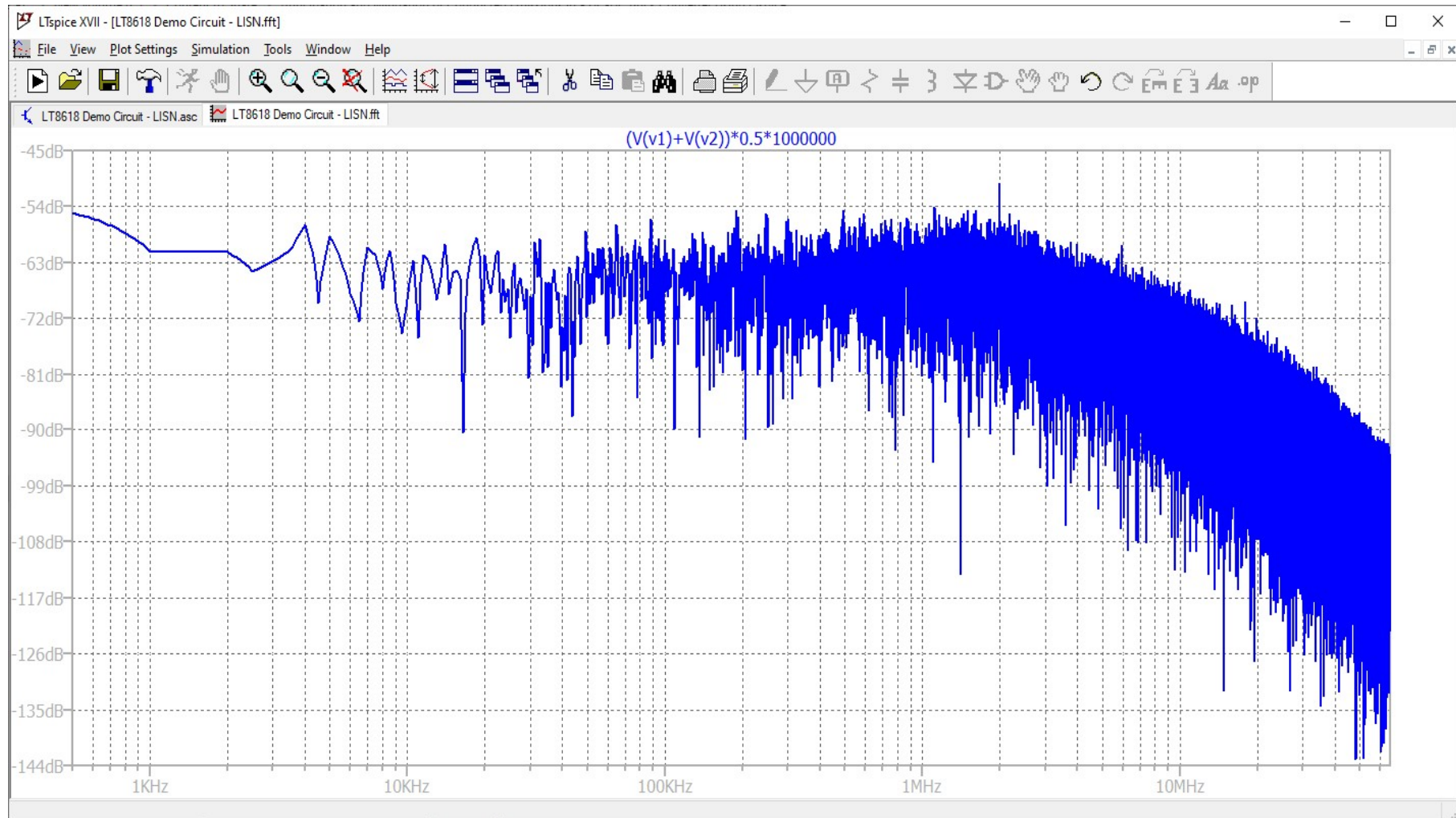


Example – Buck Converter



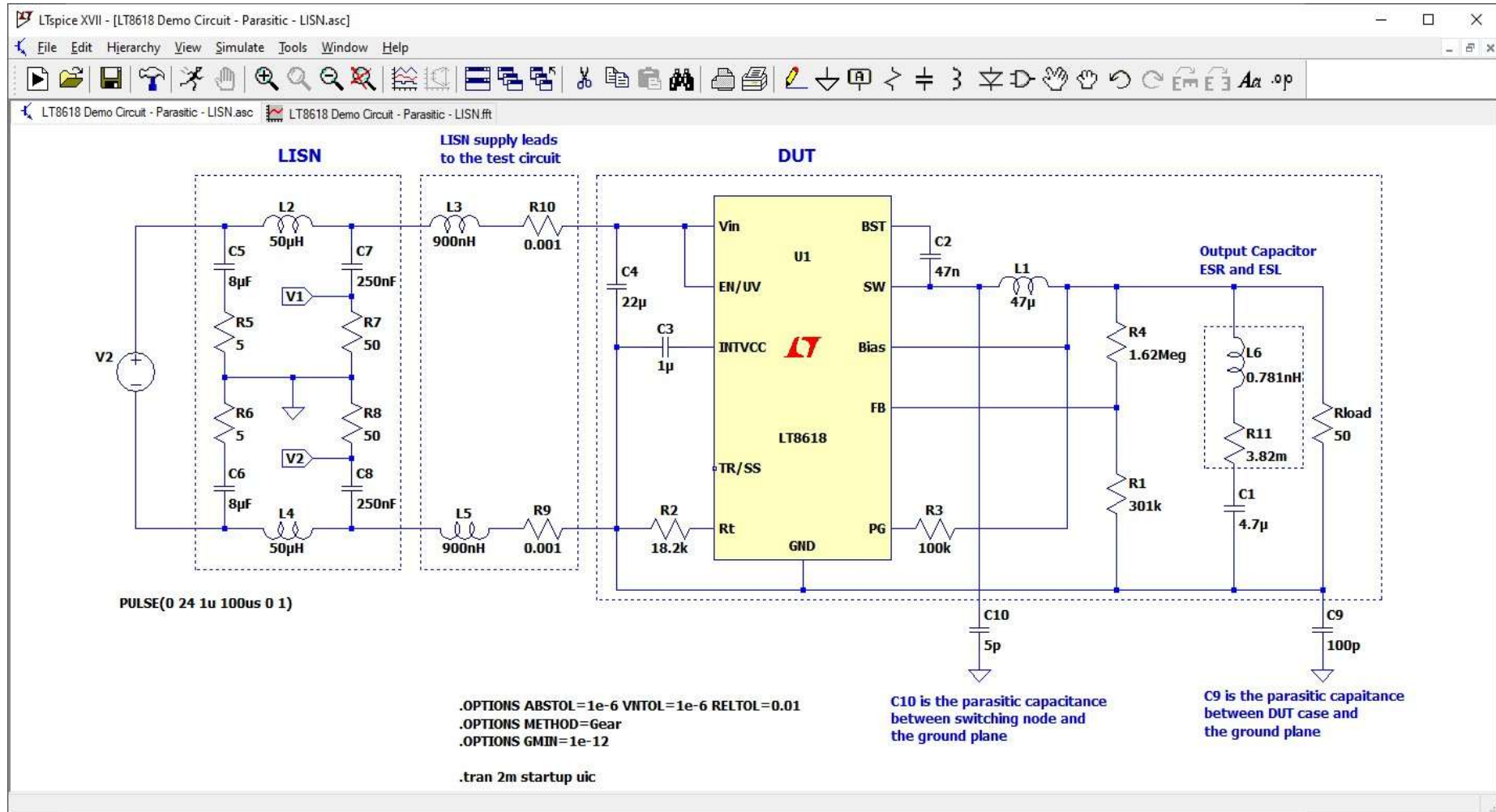
LISN Circuit according to CISPR 16-1-2:2014+A1:2017 → specifies the characteristics and performance of equipment for the measurement of radio disturbance voltages and currents

Example – Buck Converter

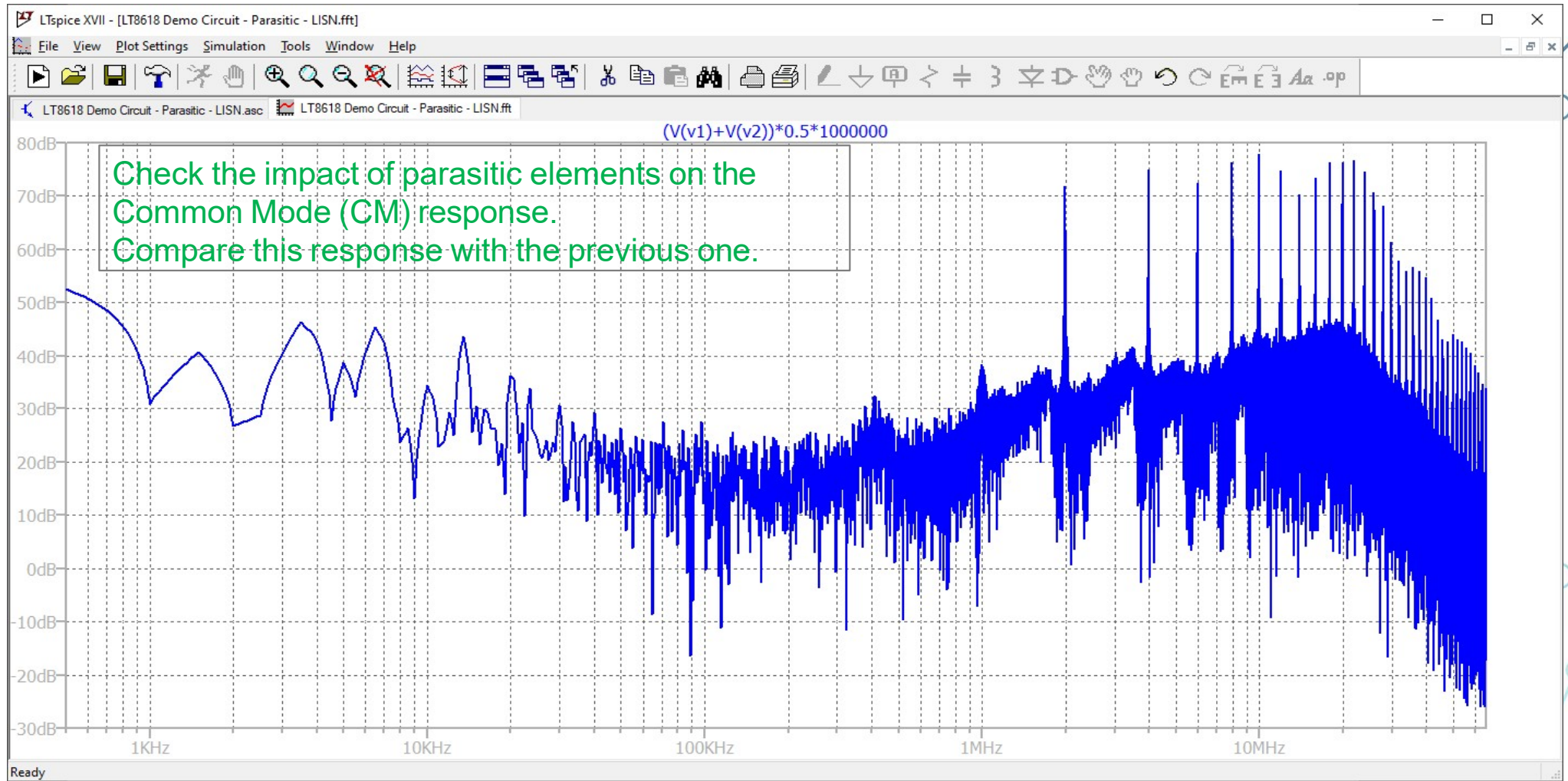


Example – Buck Converter

Adding Parasitic Elements (Stray Inductance) to Circuit Under Test

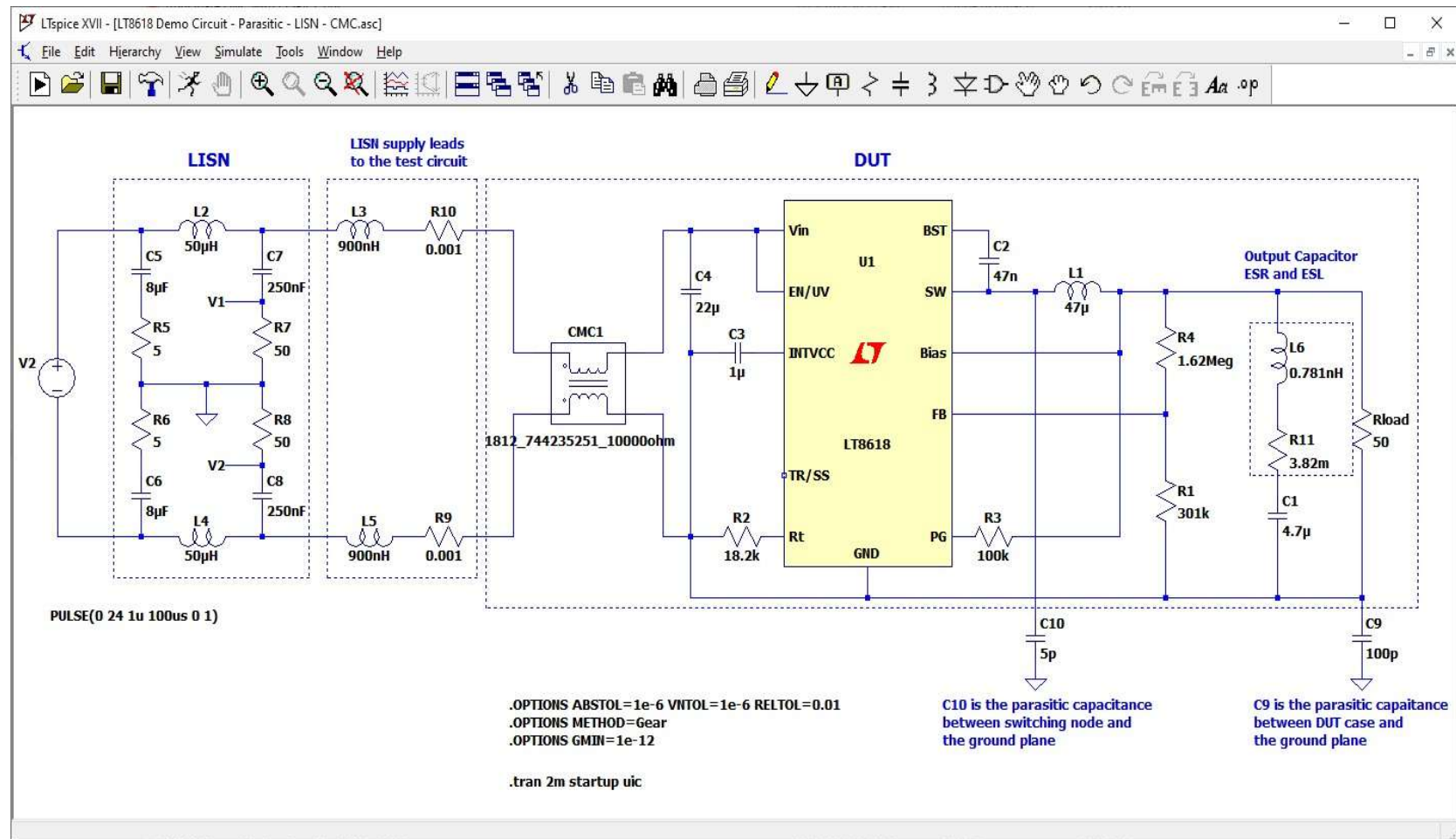


Example – Buck Converter

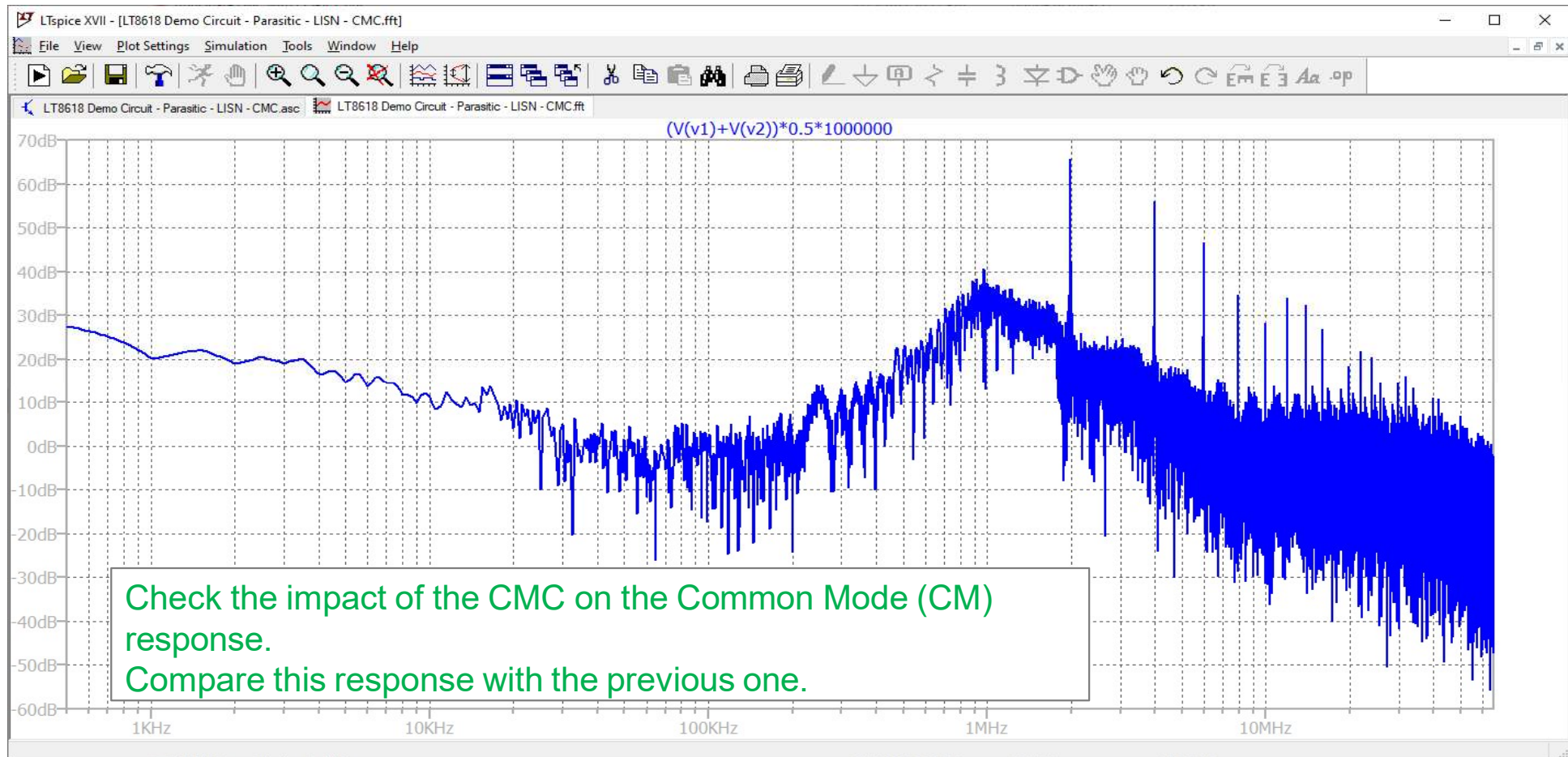


Example – Buck Converter

Adding a CMC Between the LISN and Circuit Under Test

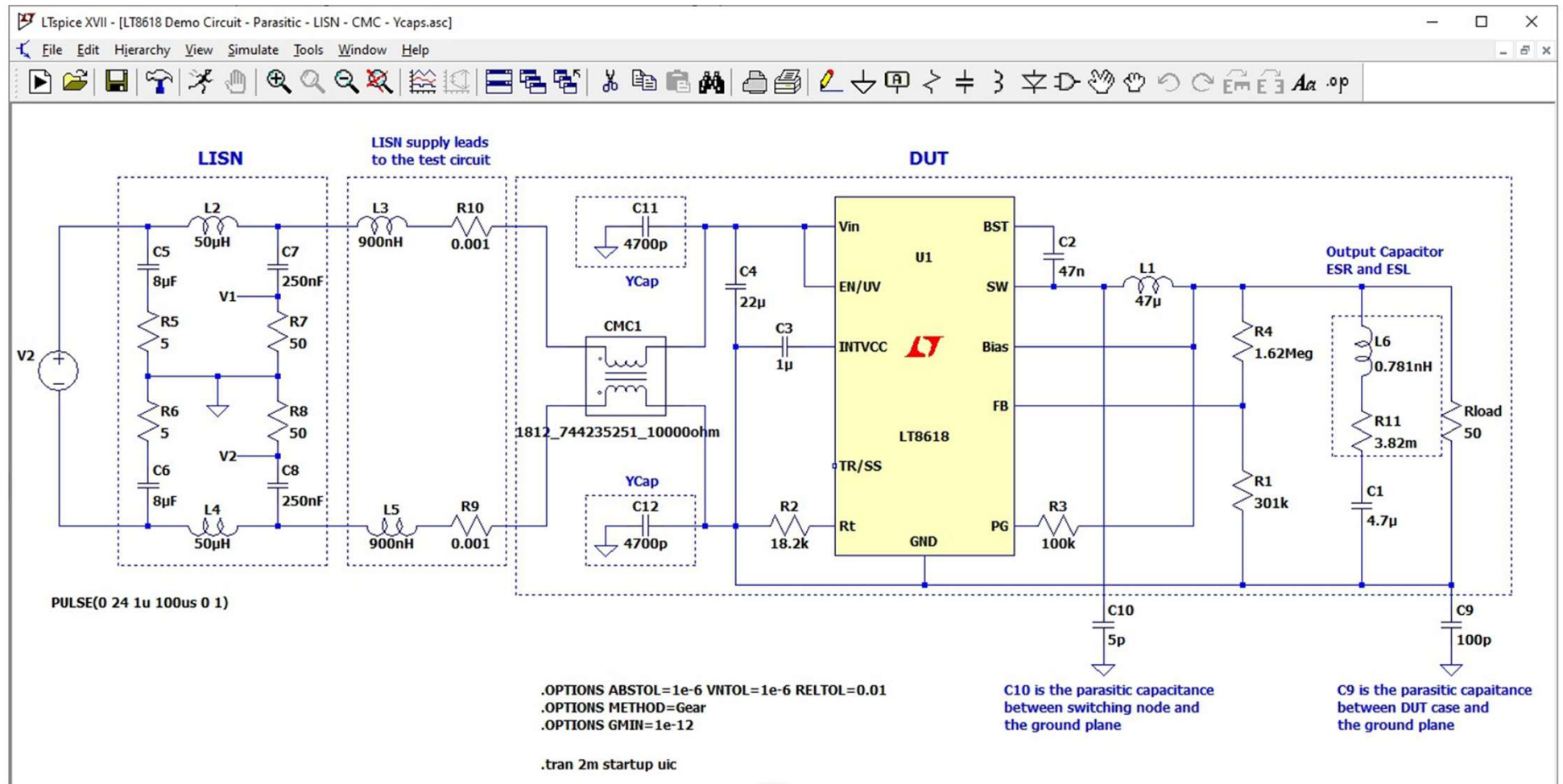


Example – Buck Converter

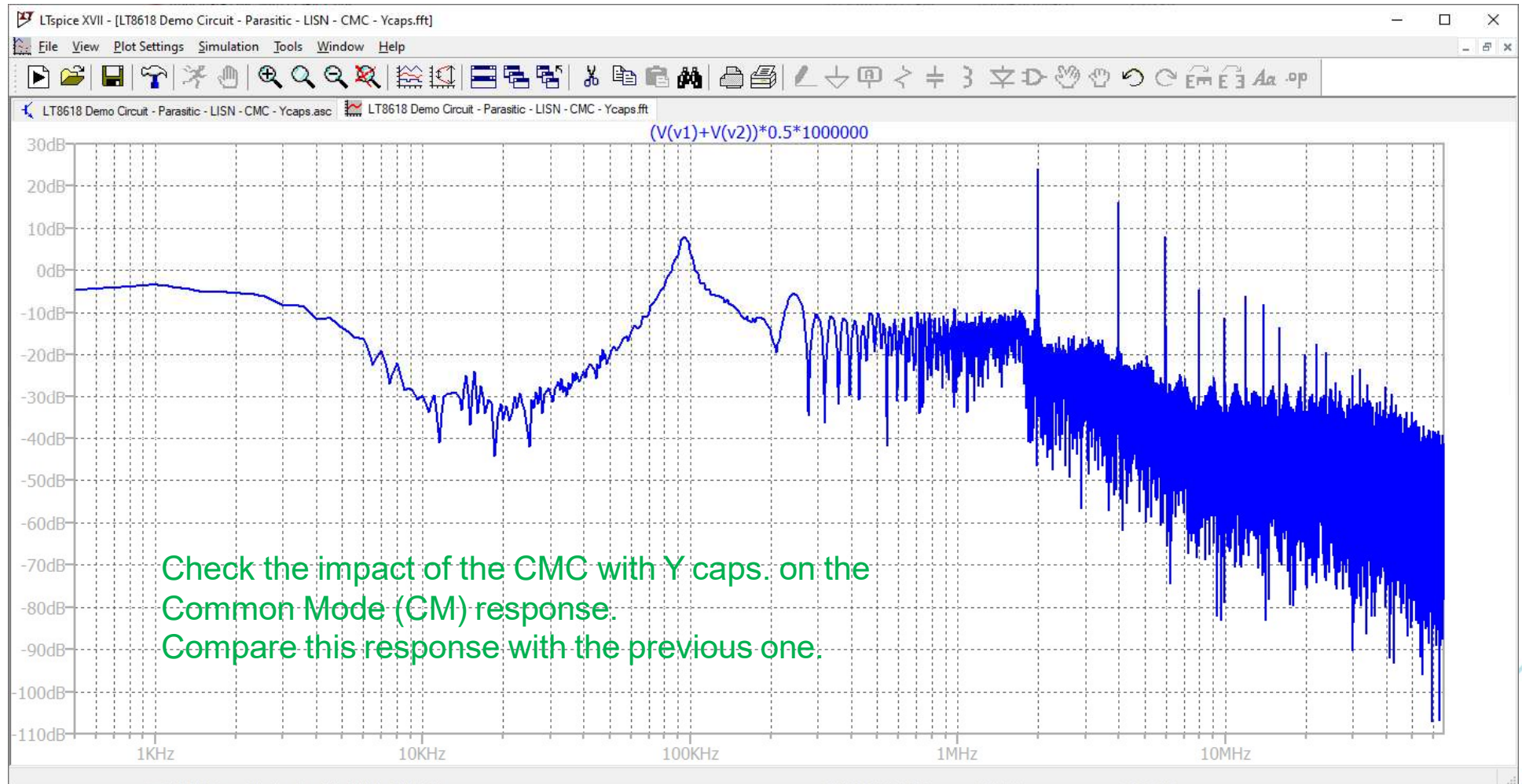


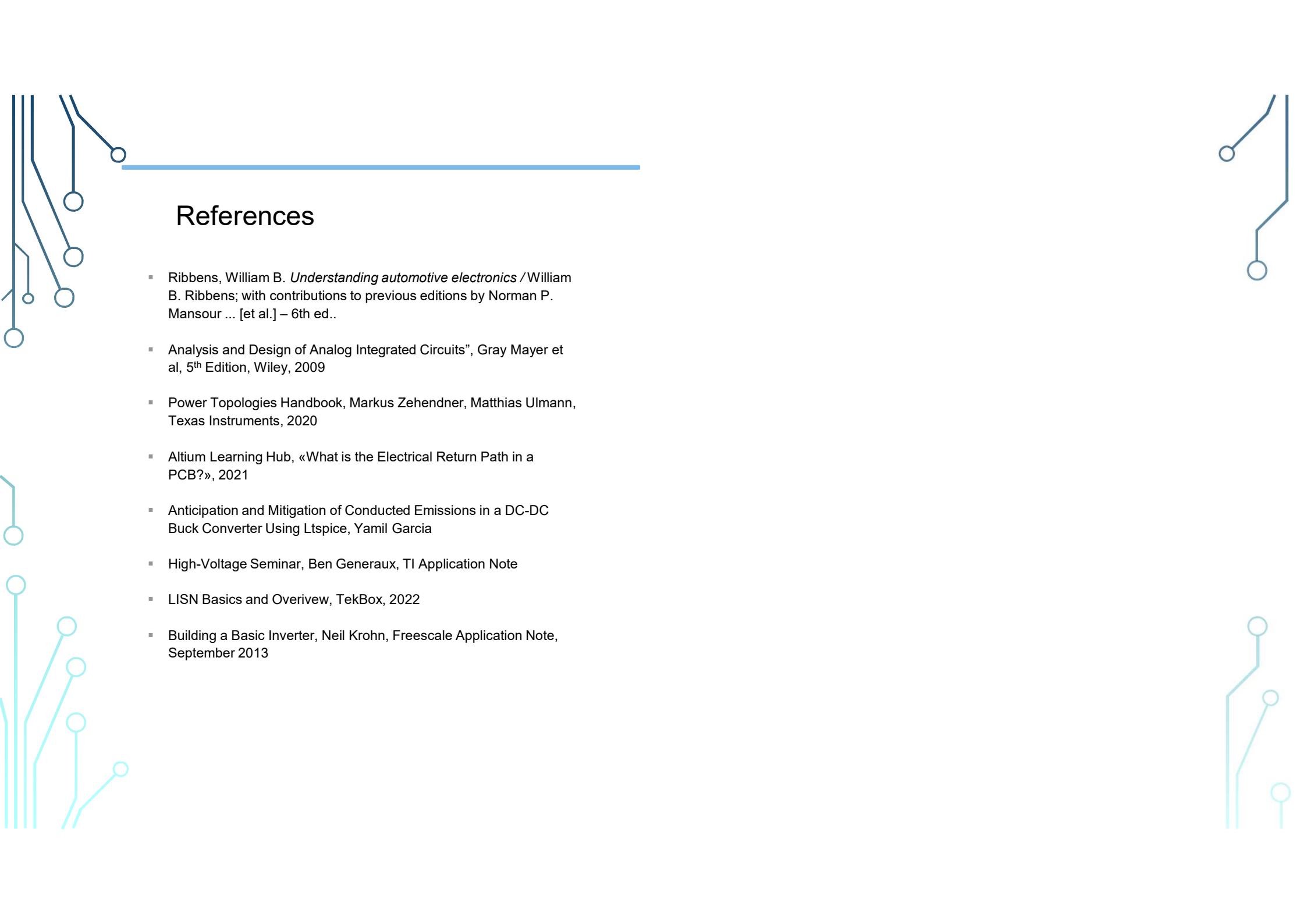
Example – Buck Converter

Adding Y Capacitors to CMC



Example – Buck Converter





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